

A novel probabilistic wind power forecasting framework integrating similar curve matching mechanism and an enhanced conditional diffusion model

Yuan Gao^a, Sile Hu^{a,b}, Yahui Chen^a, Muhammad Farhan Khan^a, Xiaolei Cheng^c,
Jiaqiang Yang^{a,*}

^a College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China

^b Inner Mongolia Power(Group) Co., Ltd., Hohhot 010020, Inner Mongolia, China

^c Inner Mongolia Daqingshan Laboratory Co., Ltd., Hohhot 010020, Inner Mongolia, China

HIGHLIGHTS

- A novel wind power probabilistic forecasting framework is introduced.
- The model integrates a Conditional Diffusion Model with Similar Curve Matching.
- The introduction of CDJN enables dynamic processing of conditional information.
- SCMM enhances wind power pattern recognition using Fast-DTW and K-Medoids.
- The framework achieves superior forecasting accuracy over existing models.

ARTICLE INFO

Keywords:

Probabilistic forecasting
Wind power prediction
Conditional diffusion model
Curve matching mechanism
Uncertainty quantification

ABSTRACT

Achieving carbon neutrality is a global priority, with wind power playing a key role in the sustainable energy transition. However, its inherent uncertainty poses challenges to power grid stability. To address this, we propose a novel probabilistic wind power forecasting framework that integrates a Similar Curve Matching Mechanism (SCMM) with an enhanced conditional diffusion model to improve uncertainty quantification. We introduce a Conditional-Denoising Joint Network (CDJN) that dynamically processes condition information, enhancing model expressiveness compared to static encoding. To account for the heterogeneity of wind power generation, SCMM leverages K-Medoids clustering and Fast-Dynamic Time Warping (Fast-DTW) to generate customized training sets, enabling the model to capture diverse wind power patterns with greater accuracy. The trained conditional diffusion model generates a large set of wind power scenarios, from which a Gated Recurrent Unit-based Autoencoder (GRUA) and Kernel Density Estimation (KDE) are applied to produce reliable probabilistic forecasts with time dependencies. This novel approach effectively enhances forecasting precision and adaptability. Case studies using real wind farm data from Ordos, Inner Mongolia, China, demonstrate that the proposed framework significantly outperforms state-of-the-art models across multiple evaluation metrics. Notably, the ablation study indicates that removing key components caused a 469.21 % increase in average loss, highlighting the critical role of each module. The results highlight its improved ability to measure wind power uncertainty and improve forecasting accuracy, hence being a promising solution to incorporate wind power into power systems more reliably.

1. Introduction

1.1. Knowledge gap and motivation

Wind power is the pillar of global decarbonization, with an installed

capacity of 1021 GW in 2023 and yearly installations growing 50 % year on year [1,2]. In spite of its rapid adoption, the inherent intermittency and variability of wind power pose significant challenges to grid stability, reliability, and operational efficiency, particularly at scale [3]. Traditional deterministic (point) forecasting techniques, although

* Corresponding author.

E-mail addresses: gaoyuan2000@ieee.org (Y. Gao), yangjiaq@zju.edu.cn (J. Yang).

<https://doi.org/10.1016/j.apenergy.2025.127031>

Received 7 March 2025; Received in revised form 5 October 2025; Accepted 3 November 2025

Available online 10 November 2025

0306-2619/© 2025 Published by Elsevier Ltd.

prevalent, do not quantify uncertainty, limiting their use to risk-aware decision-making for power system operation [4,5]. Probabilistic forecasting has therefore emerged as a necessary alternative, offering prediction intervals and quantiles to characterize uncertainty [6]. Recent advances in Artificial Intelligence (AI), particularly generative models, have enabled data-driven approaches to estimate wind power Probability Density Functions (PDFs) and generate realistic scenarios [7–10].

Among generative models, diffusion models have been prominent due to the fact that they can avoid alternatives issues: they can avoid the adversarial instability of Generative Adversarial Networks (GANs) [7], the architectural limiting assumptions of normalizing flows [9], and the approximate posterior constraints of Variational Autoencoders (VAEs) [8]. However, existing work that utilizes diffusion models in power systems has mostly addressed load forecasting [11,12], and there has been limited effort on wind power. This is a result of the unique complexities of wind generation as a consequence of nonlinear relationships between environmental variables, turbine dynamics, and surrounding conditions. Such relationships result in non-homogeneous temporal patterns that are difficult to distinguish, particularly between different time scales and operating conditions [13].

Current diffusion-based wind power methods are plagued by two primary weaknesses. Firstly, all the techniques follow static conditional injection at distinct denoising stages [14,15], which ignores dependencies across steps. Fragmented conditioning limits the model's ability to capture local dependencies within the wind power sequence, resulting in the generation of implausible scenarios [16]. Secondly, conventional training mechanisms utilize typical sets of datasets blended from a diversity of regimes without taking structural heterogeneity across power curves into consideration. This gives rise to ineffectual local-pattern learning as well as lowered predictability levels. Combating such inadequacies demands innovation within information integration conditions, as well as data preprocessing, problems this work tries to bridge.

1.2. Novelty and contributions

This work introduces a novel probabilistic forecasting framework that enhances diffusion models by two complementary innovations, overcoming the aforementioned limitations. The key contributions of this paper include:

- 1) We begin with a Conditional-Denoising Joint Network (CDJN) that learns to integrate conditional information at all diffusion steps adaptively. Unlike conventional approaches that introduce conditioning independently at each step [14,15], CDJN dynamically integrates it via a specialized network, thereby reinforcing temporal coherence throughout the denoising process. This architecture permits the model to better learn local dependencies among wind patterns, promoting both realism and statistical precision of output scenarios.
- 2) To reduce data heterogeneity, we propose a Similar Curve Matching Mechanism (SCMM). SCMM combines K-Medoids clustering with the Fast-Dynamic Time Warping (Fast-DTW) algorithm to choose training subsets of structurally similar historical wind power curves for each forecast scenario. SCMM enables the model to learn regime-specific dynamics more effectively, which are usually hidden in generic datasets [13].
- 3) A novel probabilistic forecasting framework is proposed by integrating the enhanced conditional diffusion model with SCMM. This framework effectively captures wind power uncertainty and improves forecast accuracy. Extensive validation using real-world wind farm data from the western Inner Mongolia region of China

demonstrates that the proposed approach consistently outperforms state-of-the-art models across multiple evaluation metrics, highlighting its potential for practical implementation in power system operations.

These contributions are summarized graphically in Fig. 1. To our best knowledge, this is the first effort towards synergizing dynamic diffusion conditioning with regime-aware data curation for wind power forecasting. The remainder of this paper is organized as follows: Section 2 covers related work; Section 3 presents CDJN architecture; Section 4 covers SCMM; Section 5 provides the integrated framework; Section 6 validates the approach using case studies; and Section 7 concludes the paper.

2. Related work

2.1. Traditional statistical approaches and limitations

Probabilistic wind power forecasting methods can be broadly classified into statistical and AI-based [17]. Statistical methods assume that wind power is governed by given distributions (e.g., Gaussian, Markovian) and use measured data to estimate parameters. Famous methods are Gaussian Process Models [18], Markov Chains [19], Time Series Models [20], and Copula Functions [21]. For example, Jin et al. [18] combined multiple local Gaussian Process Regression models with a finite mixture approach for improving prediction accuracy. Bounnite et al. [19] created a discrete-time Markov Chain model to model stochastic wind dynamics, while Sivhugwana et al. [20] combined wavelet transforms, AutoRegressive Integrated Moving Average (ARIMA), and Extreme Gradient Boosting (XGBoost) to simulate linear, nonlinear, and non-stationary wind speed behavior. Moradian et al. [21] integrated Archimedean Copulas with Bayesian computations to handle inter-variable correlations.

While beneficial in some contexts, these methods are faced with three primary limitations. First, they struggle to capture complex spatiotemporal and meteorological relationships inherent in wind power generation and often limit analyses to discrete power output ranges [18–20]. Second, their reliance on fixed distributions limits adaptability in the presence of fluctuating grid conditions or inhomogeneous wind regimes [21]. Third, they require significant domain expertise for parameter tuning, excluding them from generating diverse, realistic scenarios [17].

2.2. AI-based generative models: progress and challenges

To address statistical constraints, AI-driven generative models like GANs, VAEs, and Normalizing Flows (NFs) have emerged as robust tools for probabilistic forecasting. GANs leverage adversarial training between a generator and discriminator to learn data distributions [22]. Breakthroughs like Wasserstein distance optimization improved stability [23], but problems like mode collapse, vanishing gradients, and training instability persist [24]. VAEs address these issues by maximizing the variational lower bound of data likelihood. For example, Zhang et al. [24] integrated VAEs with Radial Basis Function (RBF) networks to learn spatiotemporal wind patterns, whereas Li et al. [25] incorporated self-attention mechanisms to improve nonlinear representations. Yet, VAEs impose strong posterior distributions, resulting in overlapping latent spaces and compromised accuracy [25].

NFs resolve such issues using invertible transformations that map complex distributions to simpler ones. Huang et al. [26] demonstrated the superiority of NF compared to GANs and VAEs using Unconstrained Monotonic Neural Network (UMNN) and Masked Autoencoder for

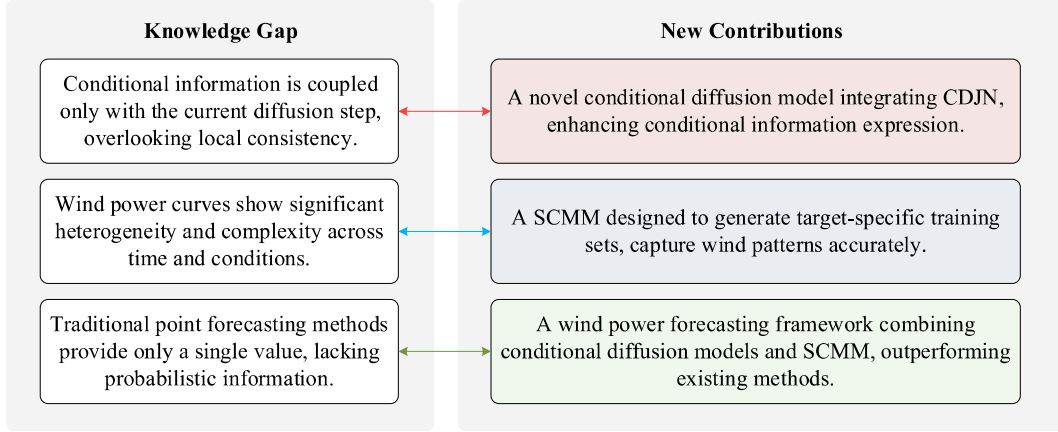


Fig. 1. Identified research gaps and key contributions of the proposed framework.

Distribution Estimation (MADE) architecture, but at increased computational expense. Less sophisticated affine autoregressive NFs have scaling issues when handling high-dimensional wind power data [27].

Recently, diffusion models have taken prominence due to their stability and versatility [28]. Diffusion models progressively transform complex data to noise and recover it through denoising, avoiding adversarial training (GANs), posterior alignment (VAEs), and invertibility constraints (NFs) [29]. In the case of power systems, diffusion models have been promising in load forecasting. Chen et al. [10] presented DiffCharge, a diffusion-based model for EV charging load curves, while Wang et al. [12] used multivariate inputs for forecasting loads. All these studies focus on load data, which have stable periodic behaviors under the constraints of human schedules [30]. Wind power has, however, dynamic meteorological conditions (turbulence and turbine dynamics) and heterogeneous time structures that are challenging to include in diffusion models [31].

One of the outstanding open problems is conditioning methods. Existing diffusion models provide conditional information (e.g., weather forecasts) at individual denoising steps [10,12], without accounting for

cross-step interactions. This piecemeal approach limits local consistency in generated scenarios, particularly for non-stable wind power curves [31]. In addition, conventional training sets mix heterogeneous wind regimes, obscuring local behavior such as cut-in/cut-out behavior or ramp events [13]. To bridge these gaps, this study introduces CDJN and SCMM, advancing diffusion models for wind power forecasting.

3. Conditional diffusion model

Traditional diffusion models operate through two primary phases: the forward diffusion process and the reverse denoising process. During forward diffusion, Gaussian noise is incrementally added to wind power scenarios until they become indistinguishable from random noise. The reverse denoising process then reconstructs wind power scenarios by progressively removing the noise [32].

Fig. 2 summarizes the workflow using block-level labels. ‘Noise addition’ corresponds to Section 3.1 (Eqs. (1)–(2)), ‘Temporal Context Learning Process’ to Section 3.2, and ‘Data restoration’ to Section 3.3 (Eqs. (3)–(5)) where the reverse transition and

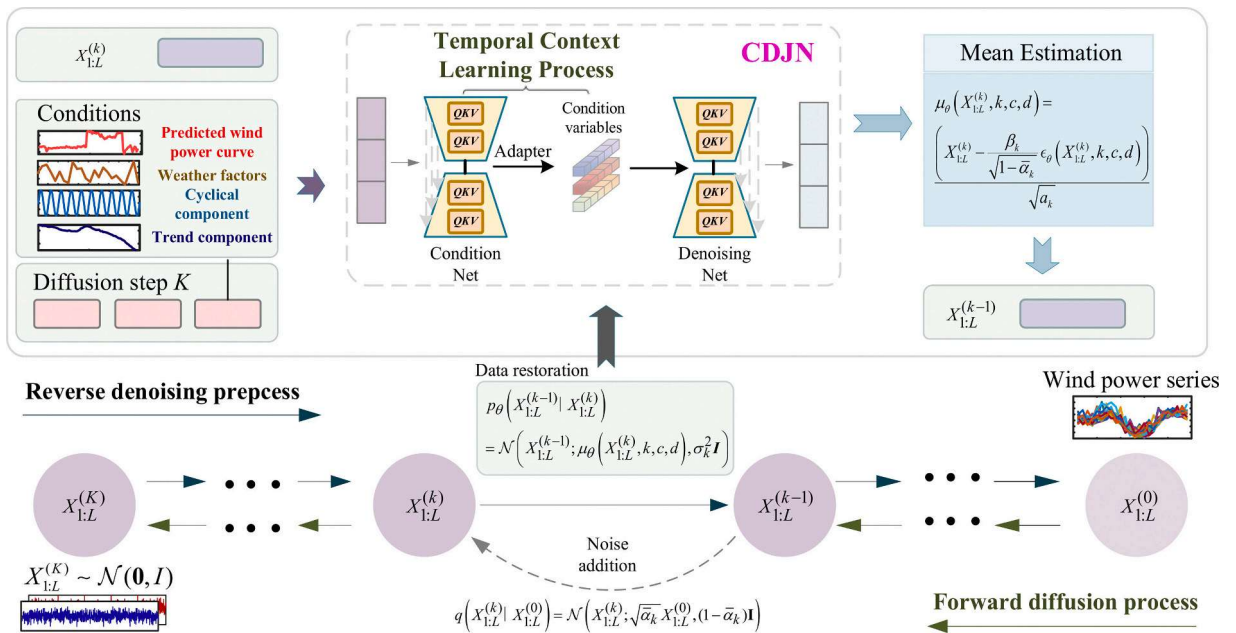


Fig. 2. Structural overview of the conditional diffusion model.

training objective are formulated.

To enable personalized wind power scenario generation, this study introduces a CDJN within the traditional diffusion model, transforming it into a Conditional Diffusion Model (Fig. 2). The CDJN consists of two core components: a Conditional Network and a Denoising Network. The Conditional Network takes the target wind power prediction sequence as input, while the Denoising Network receives the noisy sequence along with the diffusion step size. Both networks adopt a Transformer-based U-Net architecture [33] and interact through an Adapter Module.

To address wind power's non-stationary nature, the model first decomposes the sequence into trend and cyclical components. Subsequently, the day-ahead wind power forecasts, along with the meteorological variables most strongly associated as determined by the Maximal Information Coefficient (MIC) [34], are jointly incorporated to construct the conditional information component for the ensuing diffusion process. These components are then processed separately through the Conditional Network, Denoising Network, and Adapter Module, improving the model's ability to capture complex fluctuations. A key innovation introduced by CDJN is the temporal context learning process, positioned between the forward diffusion and reverse denoising phases. This additional step enables the model to better understand the temporal dependencies in wind power sequences, significantly improving the generation of realistic and personalized wind power scenarios. The details of this process will be further elaborated in the subsequent sections.

3.1. Forward diffusion process

The forward diffusion process refers to the gradual addition of Gaussian noise to the original wind power sequence $X_{1:L}^{(0)}$ over a total of K steps, until the data is completely transformed into random noise $X_{1:L}^{(K)}$. Specifically, this Markov chain process can be described as:

$$q(X_{1:L}^{(k)} | X_{1:L}^{(k-1)}) = \mathcal{N}(X_{1:L}^{(k)}; \sqrt{1 - \beta_k} X_{1:L}^{(k-1)}, \beta_k \mathbf{I}) \quad (1)$$

where $\mathbf{I}$ is identity matrix, $\mathcal{N}$ is gaussian probability distribution function, β_k is noise scaling factor, $X_{1:L}^{(k)}$ is diffusion samples, and L is sequence length. According to the Gaussian distribution superposition theorem [35], the forward diffusion process can be replaced by a single-step calculation instead of progressively adding noise, as shown in Eq. (2), significantly saving computation time.

$$q(X_{1:L}^{(k)} | X_{1:L}^{(0)}) = \mathcal{N}(X_{1:L}^{(k)}; \sqrt{\alpha_k} X_{1:L}^{(0)}, (1 - \alpha_k) \mathbf{I}) \quad (2)$$

where $\alpha_k = 1 - \beta_k$, $\bar{\alpha}_k = \prod_{i=1}^k \alpha_i$.

3.2. Temporal context learning process

In traditional diffusion models, conditional information is encoded statically, meaning the same condition is applied throughout the entire diffusion process [14]. This approach fails to account for the dynamic nature of time-series data, limiting the model's ability to capture local variations [16]. As a result, the generated data may diverge from the true distribution. To overcome these limitations, this study introduces a novel CDJN. Unlike traditional models, CDJN dynamically supplies conditional information to the Denoising Network through a dedicated Conditional Network. The Conditional Network takes as input the predicted wind power sequence, the most relevant meteorological features, and the extracted trend and cyclical components of the wind power series. At each diffusion step, the Conditional Network continuously performs temporal context learning, capturing fluctuation patterns and evolving temporal dependencies in the wind power sequence. This dynamically learned context information is transmitted in real-time to the Denoising Network, enabling it to adaptively remove noise at each

step. By leveraging this dynamic conditioning mechanism, the model generates wind power scenarios that more accurately reflect the true distribution, improving both realism and forecasting reliability.

Dynamic conditioning is realized by a stepwise context that the Conditional Network computes at every diffusion step. The network takes as inputs the predicted wind power sequence, the selected meteorological variables, and the decomposed trend and cyclical components, and produces conditioning variables that summarize short term fluctuations and long term patterns. These variables are aligned by the Adapter and delivered to the Denoising Network together with the embedded step index. This makes the reverse transition depend on the diffusion step and on the context defined in this section, which is consistent with the appearance of k , c , and d in Eqs. (4) and (5). The temporal routing follows the scheduling shown in Algorithms 1 and 2.

Specifically, in the Conditional Network, the input is the predicted wind power sequence $X_{1:L}^{(0)} \in \mathbb{R}^{R \times L}$ for the target date, where R represents the batch size and L denotes the sequence length. The sequence is divided into several non-overlapping temporal tokens via the PatchTST's Patching operation, denoted as $X_{pat} \in \mathbb{R}^{R \times P_L \times P_d}$, where P_d is the dimensional parameter of each token, P_L is the number of tokens, and the product of P_d and P_L equals L . This approach provides two key advantages. It serves in two manners. Firstly, it preserves the local patterns of the wind power sequence variation in the locality, thereby allowing the Conditional Network to extract the time series contextual dynamics successfully. Secondly, it reduces the tokens number to a minimum, lowering the Transformer's multi-head self-attention computational complexity and making it more efficient. A one-dimensional (1D) temporal convolution serves as an embedding layer to embed the input sequence $X_{pat} \in \mathbb{R}^{R \times P_L \times P_d}$ to an output sequence $X_{emb} \in \mathbb{R}^{R \times P_L \times P_d}$.

To facilitate improvement of the model's temporal dependency learning in wind power data, the Conditional Network is given not only the predicted sequence and its most relevant meteorological variables but also its trend and cyclical components. Historical data in chronological order is appended to the predicted sequence to give another temporal representation. The transformation of the selected meteorological variable matrix follows the same procedure as that of the wind power sequence, resulting in its embedding representation denoted as $\mathbf{W}_{emb} \in \mathbb{R}^{R \times P_L \times P_d}$. In addition, a sophisticated Empirical Mode Decomposition (EMD) and Sample Entropy Decomposition [36] is applied to decompose the wind power time series into trend and cyclical components. EMD offers an adaptive, data-driven representation that is well suited to the nonstationary and intermittent characteristics of wind power. By extracting intrinsic mode functions directly from the data, EMD preserves sharp ramps and regime shifts without prespecifying a seasonal period or a basis. The entropy-based screening then retains informative modes and suppresses noisy or redundant components, mitigating leakage of high-frequency variation into the trend. In contrast, Seasonal-Trend decomposition using Loess (STL) assumes an additive trend-seasonal structure with a fixed seasonal period [37], and wavelet decompositions depend on the choice of mother wavelet and scale grid [38]. Under strongly time-varying dynamics these assumptions can cause basis mismatch and boundary artifacts. For these reasons we adopt EMD with entropy-guided selection in this study.

The decomposition allows the model to disentangle long-term trends from short-term fluctuations, making it more capable of generating more accurate wind power scenarios. Subsequently, a 1D temporal convolution is applied to embed them, producing trend prompt vector $T_{emb} \in \mathbb{R}^{R \times P_{pro}}$ and cyclical prompt vector $C_{emb} \in \mathbb{R}^{R \times P_{pro}}$, with P_{pro} being the embedding dimension hyperparameter.

During the context learning process, the Conditional Network processes the input data using the multi-head self-attention mechanism of the Transformer to capture temporal dependencies between tokens. By analyzing historical fluctuation patterns across multiple time scales, the network extracts a comprehensive set of temporal representations that

reflect the evolving dynamics of wind power sequences. These temporal representations are then processed through an adaptive module to generate a set of conditional variables. These variables, acting as dynamic context information, are transmitted to different modules within the Denoising Network. By integrating this context-driven guidance, the Denoising Network can more effectively remove noise at each diffusion step, ensuring that the generated wind power scenarios align more accurately with real-world distribution patterns.

3.3. Reverse denoising process

The reverse denoising process refers to the process of removing noise from random noise $X_{1:L}^{(0)}$ to recover the original data $X_{1:L}^{(0)}$. This process can be modeled using a Markov chain with learnable transitions:

$$p_{\theta}(X_{1:L}^{(0:K)}) = p(X_{1:L}^{(K)}) \prod_{k=1}^K p_{\theta}(X_{1:L}^{(k-1)} | X_{1:L}^{(k)}) \quad (3)$$

In the Denoising Network, the input is the noisy sequence $X_{1:L}^{(k)} \in \mathbb{R}^{R \times L}$, which is the predicted wind power sequence with added noise. Through the same temporal block partitioning operation used in the Conditional Network, this sequence is divided into $X_{pat}^{(k)} \in \mathbb{R}^{R \times P_L \times P_d}$. However, in this process, a fully connected layer is applied to embed it into $X_{emb}^{(k)} \in \mathbb{R}^{R \times P_L \times P_d}$. At the same time, since the reverse denoising process requires the diffusion step k as guidance, the model similarly embeds this step through a fully connected layer to obtain k_{emb} . To further improve denoising accuracy, the model incorporates both the real-time wind power sequence context information captured by the Conditional Network, meteorological features, as well as the trend and cyclical components of the wind power sequence as auxiliary inputs. This

integration enhances the model's ability to distinguish between short-term fluctuations and long-term patterns, ensuring a more precise reverse denoising process. The optimized denoising process is structured as follows:

$$p_{\theta}(X_{1:L}^{(k-1)} | X_{1:L}^{(k)}) = \mathcal{N}(X_{1:L}^{(k-1)}; \mu_{\theta}(X_{1:L}^{(k)}, k, c, d), \sigma_k^2 I). \quad (4)$$

where $\mu_{\theta}(X_{1:L}^{(k)}, k, c, d)$ is the predicted results of the conditional neural network, c is the conditional information, d is the auxiliary information, θ is the parameters learned by the reverse Denoising Network, and σ_k^2 is the hyperparameter for the fixed variance. In our implementation the mean parameter μ_{θ} in Eq. (4) should be read as a function of the noisy input, the step index, and the conditioning variables provided by Section 3.2, which matches the notation $\mu_{\theta}(X_{1:L}^{(k)}, k, c, d)$ used in Eq. (4), and the training objective in Eq. (5) depends on the same variables.

To obtain the optimal $\mu_{\theta}(X_{1:L}^{(k)}, k, c, d)$, the model training is carried out by maximizing the Variational Lower Bound of the Probability (ELBO) [39]. Through theoretical derivation and experimental analysis, the training objective l_{diff} can be simplified as follows:

$$l_{diff}(\theta) = E_{X_{1:L}^{(0)}, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\alpha_k} X_{1:L}^{(0)} + \sqrt{1 - \alpha_k} \epsilon, k, c, d \right) \right\|^2 \right] \quad (5)$$

where ϵ_{θ} represents the predicted noise, and ϵ is the random noise sampled from a standard Gaussian distribution. The conditioning variables are formed by the Conditional Network described in Section 3.2 and are supplied to the Denoising Network at each step during training and inference. The model training and inference processes are shown in Algorithm 1 and Algorithm 2, respectively, where U is a uniform

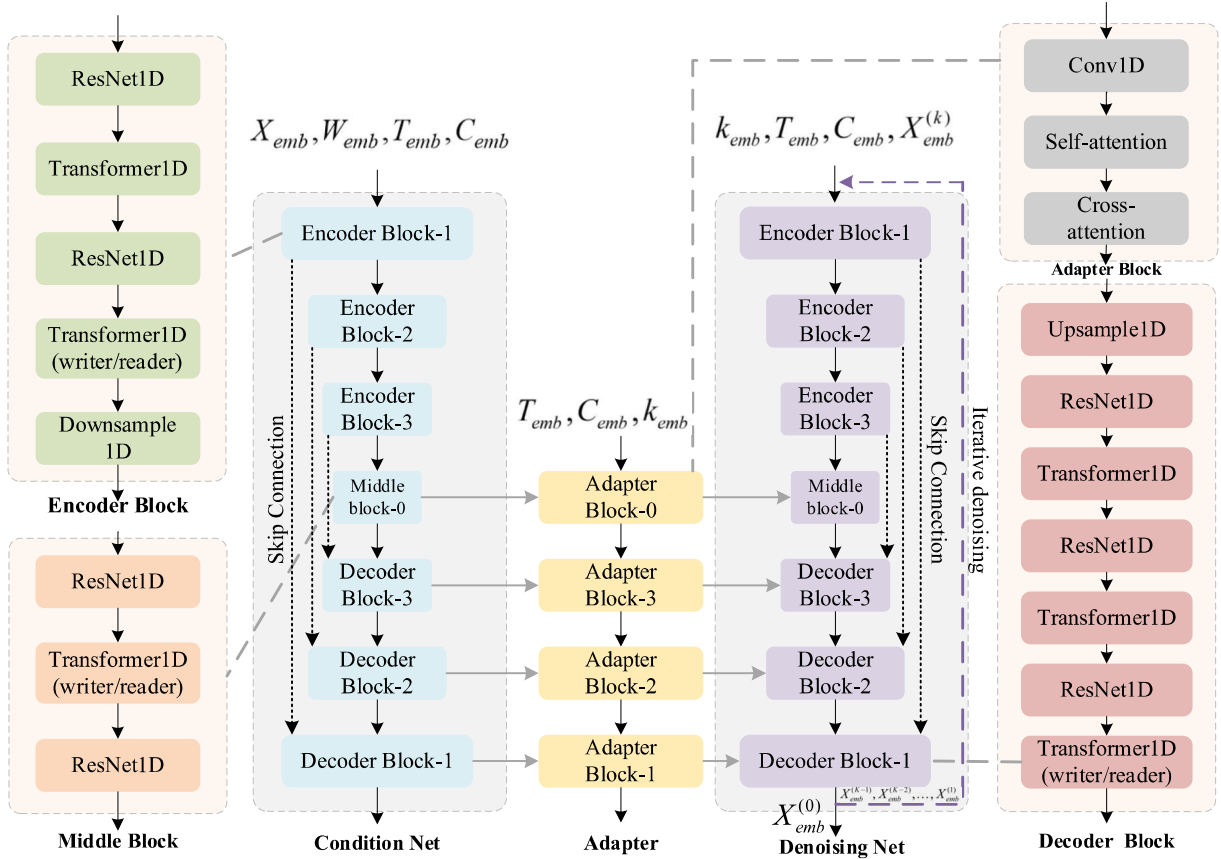


Fig. 3. Architecture of the CDJN.

distribution.

module. Encoder Block 1 and Decoder Block 1 have a feature dimension of $P_d/2$ and a temporal dimension of P_L , connected via a skip connection.

Algorithm 1: Training Procedure for the Conditional Diffusion Model.

Step 1: Initialization of diffusion step size K , maximum training epoch ξ , training batch R , Adam optimizer hyperparameters β_1 , β_2 , Adam optimizer learning rate l_r .

Step 2: Initialization of network parameters θ .

Step 3: Network training

for training epoch ξ do

 for batch size R do

 # Retrieve historical wind power data x_r

$x^{(0)} = x_r, r \in [1, R]$

 # Randomly select diffusion steps and generate noise samples

$k = U(1, K)$

$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

 # Calculate noise data based on noise sample ϵ and diffusion step k

$X_{1:L}^{(k)} = \sqrt{\alpha_k} X_{1:L}^{(0)} + \sqrt{1 - \alpha_k} \epsilon$

 # Data Storage

 Save the corresponding $k, x^{(0)}, \epsilon$, and assign to $x_r = x^{(k)}$

 # Predict noise data using the CDJN

$\epsilon_\theta(\sqrt{\alpha_k} X_{1:L}^{(0)} + \sqrt{1 - \alpha_k} \epsilon, k, c, d)$

 # Update the network using the gradient descent method

$\theta \leftarrow \text{adam}(\nabla_\theta \|\epsilon - \epsilon_\theta\|^2, \theta, l_r, \beta_1, \beta_2)$

 end for

end for

Output θ .

Algorithm 2: Inference Process for Generating Wind Power Scenarios Using the Conditional Diffusion Model.

Step 1: Initialization of diffusion steps K .

Step 2: Denoising process

for training batch R do

 for $k = K, \dots, 1$ do

 # Retrieve the initial noise image

$X_{1:L}^{(k)} = x_r, r \in [1, R]$

 # Randomly generate noise sample z

 if $k > 1$

$z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

 else

$z = \mathbf{0}$

 # Predict the noise added at diffusion step k

$\epsilon_\theta(X_{1:L}^{(k)}, k, c, d)$

 # Calculate the data at diffusion step $k-1$

$X_{1:L}^{(k-1)} = \frac{1}{\sqrt{\alpha_k}} \left(X_{1:L}^{(k)} - \frac{\beta_k}{\sqrt{1 - \alpha_k}} \epsilon_\theta \right) + \sqrt{\sigma_k^2} \cdot z$

 end for

end for

Output $X_{1:L}^{(0)}$ of the r -th batch.

3.4. Conditional diffusion model network architecture design

In the proposed Conditional Diffusion Model, the CDJN serves as the core component, ensuring effective integration of both local details and contextual information. To achieve this, both the Conditional Network and the Denoising Network are designed using a Transformer-based U-Net architecture. This structure consists of an encoder, a decoder, and an intermediate module, as illustrated in Fig. 3.

As shown in Fig. 3, the CDJN share a three-stage encoder-decoder structure, with the encoder and decoder linked through a central

Similarly, Encoder Block 2 and Decoder Block 2 have a feature dimension of $P_d/4$ and a temporal dimension of $2P_L$, while Encoder Block 3 and Decoder Block 3 have a feature dimension of $P_d/8$ and a temporal dimension of $4P_L$. Each encoder block concludes with a Downsample1D layer, while each decoder block begins with an Upsample1D layer.

In the encoder, deeper layers progressively enhance the resolution of the temporal sequence dimension, allowing the model to first extract global patterns at lower resolutions before refining local details at higher resolutions. Conversely, the decoder gradually decreases the resolution, facilitating the transformation from high-resolution local

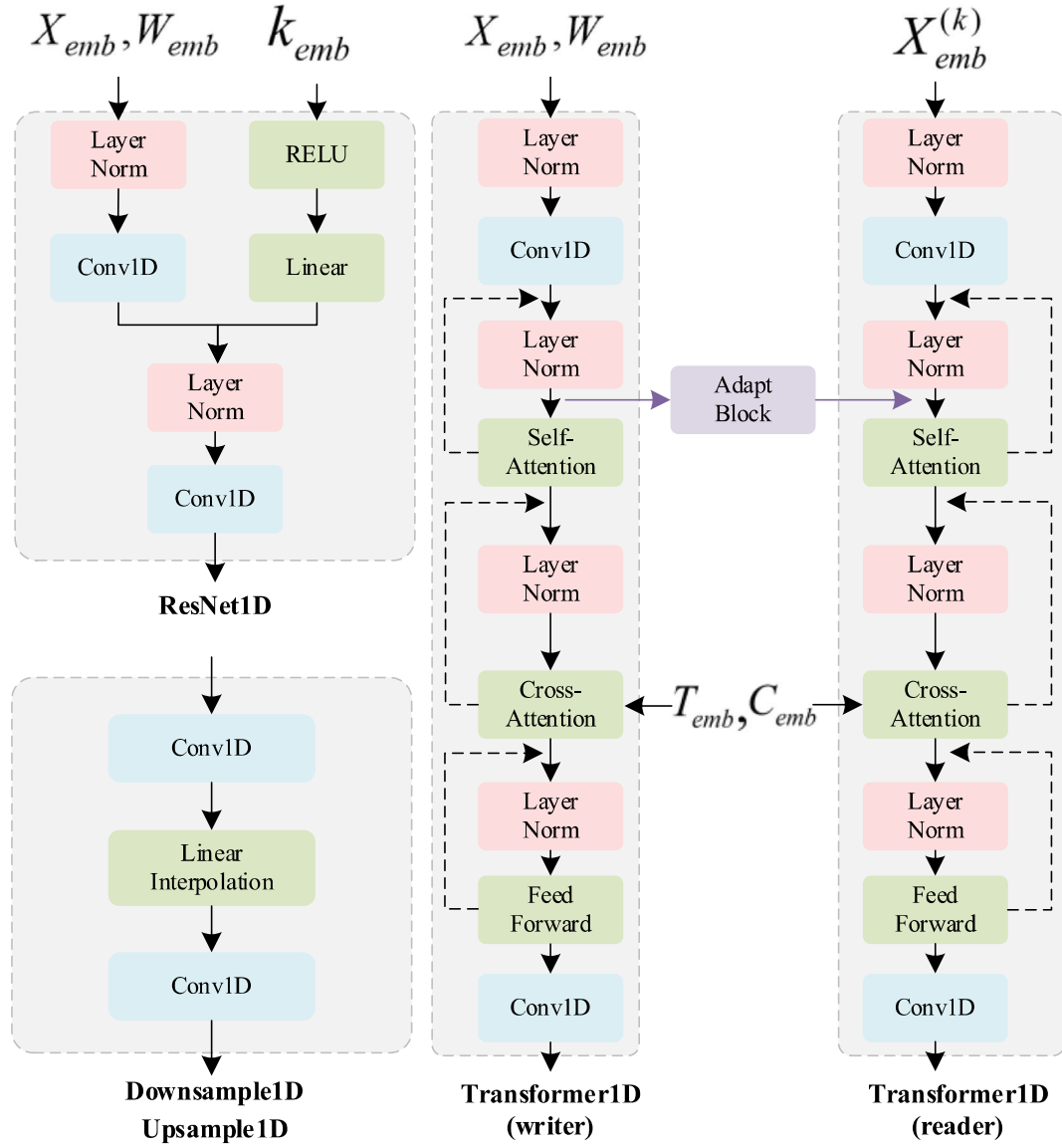


Fig. 4. Detailed module configuration of the conditional diffusion model.

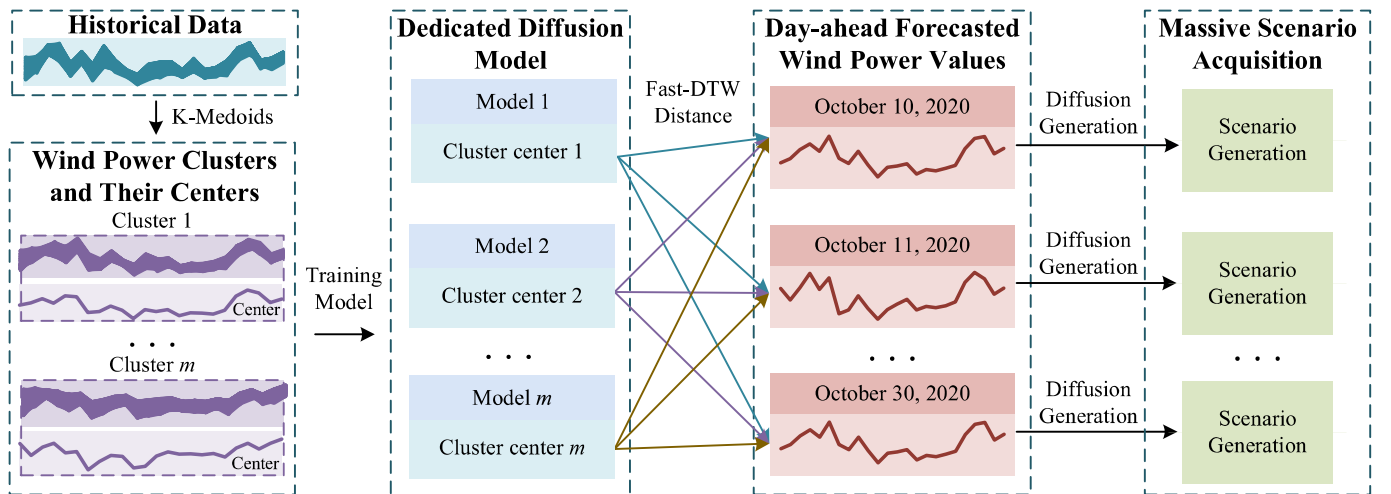


Fig. 5. Overall process of SCMM.

details to low-resolution global representations. The Conditional Network and Denoising Network are interconnected through the central module and decoder blocks, allowing efficient information transfer. Additionally, trend and cyclical components are integrated into the Conditional Network, Denoising Network, and Adapter Module via a cross-attention mechanism, enabling the model to perceive underlying temporal patterns and optimize the generation process.

The core architecture of the CDJN consists of ResNet1D, Down-sample1D, Upsample1D, and Transformer1D modules, as detailed in Fig. 4. The Transformer1D module is further divided into two sub-modules: Writer and Reader, corresponding to the Conditional Network and Denoising Network, respectively. The Writer module utilizes the self-attention mechanism to capture dependencies between tokens, facilitating global modeling of the wind power sequence. Meanwhile, the Reader module refines the denoising process by leveraging interactions between the wind power trend's cyclical pattern and the noisy sequence, ensuring more accurate and realistic scenario generation.

4. Similar curve matching mechanism

In the reverse denoising process of the conditional diffusion model, noise is incrementally removed to reconstruct the original wind power data. However, due to the inherent heterogeneity and complexity of wind power across different time periods and conditions, this iterative recovery process presents significant challenges. The model may struggle to accurately capture the true fluctuation patterns of wind power, leading to deviations in the generated scenarios. To address this issue, we introduce the SCMM to enhance the model's ability to learn and represent wind power variations more effectively.

SCMM begins by applying K-Medoids clustering to group historical wind power curves into multiple clusters (suppose there are m clusters), extracting the centroid curve for each cluster [40]. Since curves within the same cluster exhibit high similarity, this clustering process enables the model to focus on learning the specific variation patterns associated with each category of wind power fluctuations. Next, a dedicated conditional diffusion model is trained separately for each cluster, using a training dataset specific to that cluster. To ensure feasibility, the number of clusters is restricted by validity indices so that only a small set of representative regimes is defined. All diffusion models adopt the same architecture and training configuration, allowing parallel training and efficient reuse of computational resources. During operation, only the model corresponding to the closest regime is invoked, thereby keeping the framework computationally practical. This tailored modeling approach significantly improves the quality of scenario generation, particularly when wind power fluctuations exhibit distinct categorical characteristics. The impact is especially pronounced when power variation patterns align with clearly defined classification traits. Beyond improving scenario generation, the clustered models also enhance interpretability. Since each model corresponds to a specific wind power fluctuation pattern, it can be mapped to a particular wind power generation regime associated with a defined wind speed range. To select the most appropriate model for a given forecasting target, we compare the predicted wind power values for the target date against all cluster centroids using the Fast-DTW algorithm [41]. The conditional diffusion model associated with the closest matching centroid is then chosen for scenario generation. The overall process of SCMM is described in Fig. 5, and the implementation details of K-Medoids clustering and Fast-DTW distance calculations will be elaborated in the following sections.

Fig. 5 summarizes how clustering, model selection, and temporal context operate within SCMM, while Sections 3.2 and 3.3 provide the mathematical formulation of the conditioning pathway and the reverse transition.

4.1. K-Medoids clustering

K-Medoids clustering is employed to identify representative cluster

centroids from a large set of wind power curves. Unlike K-Means, which computes synthetic mean centroids, K-Medoids selects actual samples from the dataset as cluster centers [42]. This ensures that the structural features of wind power curves, including sharp ramps, non-Gaussian fluctuations, and irregular turbine responses, are preserved. Such fidelity is critical for SCMM, as it allows the conditional diffusion model to be trained on physically meaningful regimes rather than averaged representations. Furthermore, K-Medoids demonstrates greater robustness to outliers than K-Means, which is particularly important in wind power datasets containing extreme or rare events. For these reasons, K-Medoids provides a more reliable foundation for scenario generation in SCMM. Given that this method is widely recognized, its detailed mathematical formulation is not reiterated in this section.

4.2. Fast-DTW distance

DTW distance is a technique used to measure the similarity between two-time series by dynamically aligning them along the time axis. It computes the optimal warping path between sequences to minimize alignment distortions, providing a robust similarity measure even when time series exhibit phase shifts or varying temporal scales [43]. Given two time series, I^a and I^b , the distance matrix $D_{T \times T}$ is constructed, where each element represents the pairwise distance computed using the formula in Eq. (6). The Fast-DTW algorithm, an optimized version of DTW, is employed to improve computational efficiency while maintaining accurate similarity measurements. In this case, $I^a = [I_1^a, I_2^a, \dots, I_T^a]$ represents the wind power matrix for the target date, and $I^b = [I_1^b, I_2^b, \dots, I_T^b]$ represents the matrix of the centroid curve for a particular cluster.

$$D(\phi, \varphi) = \sqrt{(I_\phi^a - I_\varphi^b)^2} \quad 1 \leq \phi \leq T, 1 \leq \varphi \leq T \quad (6)$$

where $D(\phi, \varphi)$ is the DTW distance between points I_ϕ^a and I_φ^b , T is the period of the sequence. The goal of DTW is to find the optimal warping path $W = (w_1, w_2, \dots, w_O)$, such that the cumulative distance from the starting point $w_1 = (1, 1)$ to the endpoint $w_O = (\phi, \varphi)$ is minimized. Here, O is the total number of point pairs on the path W .

$$I_{DTW}(W) = \min_W \sum_{o=1}^O D(w_o) \quad (7)$$

where $I_{DTW}(W)$ is the DTW distance, and w_o represents the path element at the o -th position in the matrix, which typically needs to satisfy the requirements of boundary conditions, continuity, and monotonicity. For Eq. (7), the cumulative distance matrix is constructed and solved using dynamic programming. However, as the length of the time series increases, the computational complexity of the DTW algorithm becomes a major limitation, reducing efficiency. To address this, we employ the Fast-DTW algorithm, as proposed in [39]. This method significantly enhances computational efficiency by restricting the search space within the distance matrix, ensuring a faster yet accurate similarity calculation.

5. Wind power probability forecasting framework

This study introduces a novel wind power probabilistic forecasting framework by integrating the Conditional Diffusion Model with the SCMM, as illustrated in Fig. 6.

The framework follows a structured process:

Step 1: Collect historical wind power curves along with the wind power day-ahead prediction curve for the target date, as provided by the system operator.

Step 2: The day-ahead prediction curve is concatenated with the historical wind power sequence, after which Empirical Mode Decomposition with entropy-based screening is applied to extract trend and cyclical components (see Section 3.2). These components are further

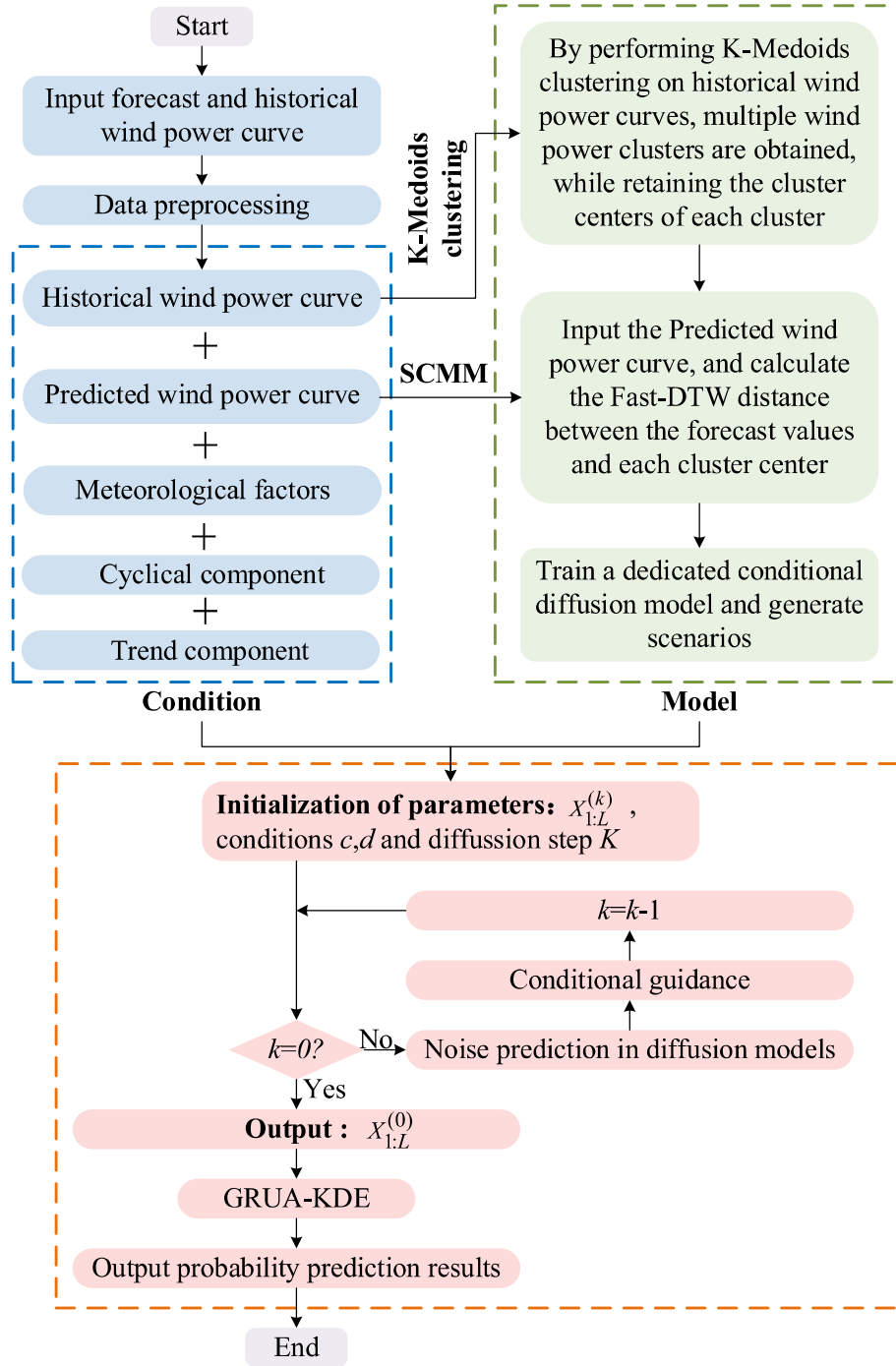


Fig. 6. Wind power probabilistic forecasting framework workflow.

enriched by incorporating meteorological features to enhance the overall representation of the input data.

Step 3: Utilize K-Medoids clustering to categorize historical wind power curves into multiple clusters, retaining the centroid curve for each cluster.

Step 4: Compute the Fast-DTW distance between the predicted curve for the target date and all centroid curves, selecting the cluster whose centroid best matches the predicted trend.

Step 5: Construct a dedicated training set based on the selected cluster and use multiple noise and condition parameters to guide the Conditional Diffusion Model, generating a diverse set of realistic wind power scenarios.

Step 6: Combine a Gated Recurrent Unit-based Autoencoder (GRUA)

with KDE to estimate the PDF of the generated wind-power scenario set. The GRUA captures sequence dependencies and temporal correlations across time steps, while KDE provides nonparametric density estimation; together they yield a probabilistic wind-power forecast for the target date (The specific details of GRUA and KDE are discussed in detail in the following sections).

5.1. Construction of GRUA

After generating a large number of wind power scenarios using the Conditional Diffusion Model, directly applying KDE for PDF estimation may fail to effectively capture the temporal dependencies present in the generated scenarios [44]. This limitation arises because KDE, as a non-

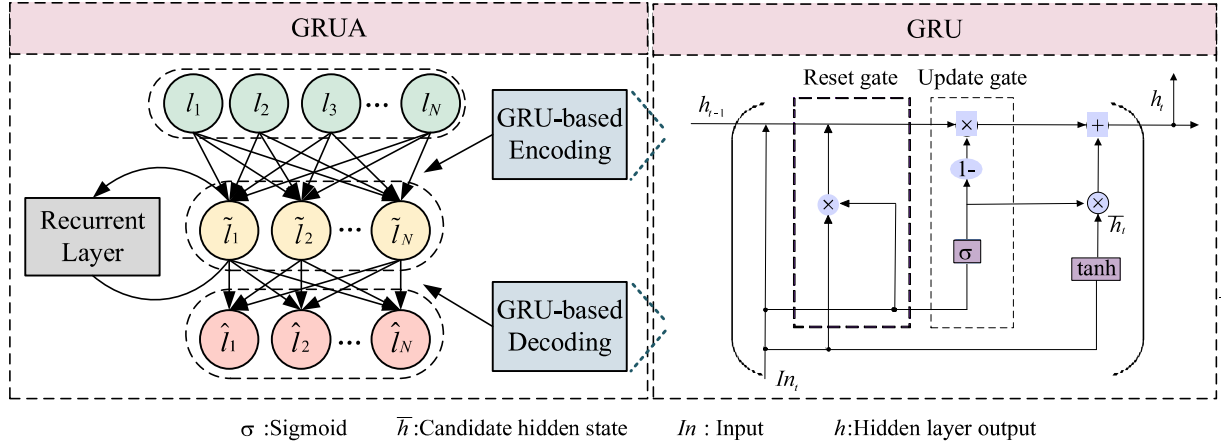


Fig. 7. The structure of the GRUA model.

parametric estimation method, independently operates at each time step, focusing solely on the marginal distribution characteristics of individual time steps. Consequently, it overlooks the continuity and correlations of wind power output across the time dimension, leading to a loss of temporal dependency information. To address this issue, this study introduces a GRUA model to learn and extract the sequence dependencies across time steps, providing feature representations that incorporate temporal correlations for the subsequent KDE process.

The structure of the GRUA model is illustrated in Fig. 7, which incorporates both the GRU and autoencoder modules. While the autoencoder, as an unsupervised learning model, is capable of compressing input data into latent space and reconstructing it, it struggles to capture temporal dependencies when processing time-series data such as wind power output [45]. The GRU, a specialized recurrent neural network for sequence processing, effectively models long-term dependencies through its update and reset gate mechanisms [46]. By integrating the GRU into the autoencoder, the encoder retains temporal features while compressing the data, allowing the latent space representation to encapsulate both the content of the data and its temporal dynamics. This integration enhances the model's ability to capture and represent time-series dependencies.

Specifically, encoding and decoding are the two key steps in an autoencoder. Given N wind power scenarios $\mathbf{l}^c = [l_1^c, l_2^c, \dots, l_N^c]$ generated by the Conditional Diffusion Model, the encoding step is represented as

shown in Eq. (8).

$$\tilde{\mathbf{l}} = \mathbf{f}_1^s(\mathbf{W}_1^s \cdot \mathbf{l} + \mathbf{b}_1^s) \quad (8)$$

where $\tilde{\mathbf{l}}$ is the encoded wind power scenarios, and $\mathbf{W}_1^s$, $\mathbf{b}_1^s$, and $\mathbf{f}_1^s$ are the weight terms, bias terms, and activation terms, respectively. The decoding step of the autoencoder is given by Eq. (9).

$$\hat{\mathbf{l}} = \mathbf{f}_2^s(\mathbf{W}_2^s \cdot \tilde{\mathbf{l}} + \mathbf{b}_2^s) \quad (9)$$

where the reconstructed wind power scenarios are denoted by $\hat{\mathbf{l}}$, and $\mathbf{W}_2^s$, $\mathbf{b}_2^s$, and $\mathbf{f}_2^s$ are the weight terms, bias terms, and activation terms, respectively. The basic structure of the GRU is well-established, as described in [47].

5.2. KDE steps

The PDF of the reconstructed wind power scenario $\hat{\mathbf{l}}^c$ is constructed using the KDE method [48]. The estimator is given by:

$$\hat{f}(\mathbf{l}^c) = \frac{1}{N} \sum_{n=1}^N v\left(\frac{\mathbf{l}^c - \hat{\mathbf{l}}_n^c}{\mathcal{H}}\right) \quad (10)$$

where $\hat{f}(\mathbf{l}^c)$ is the estimator of the wind power sequence PDF, $\hat{\mathbf{l}}^c$ is the

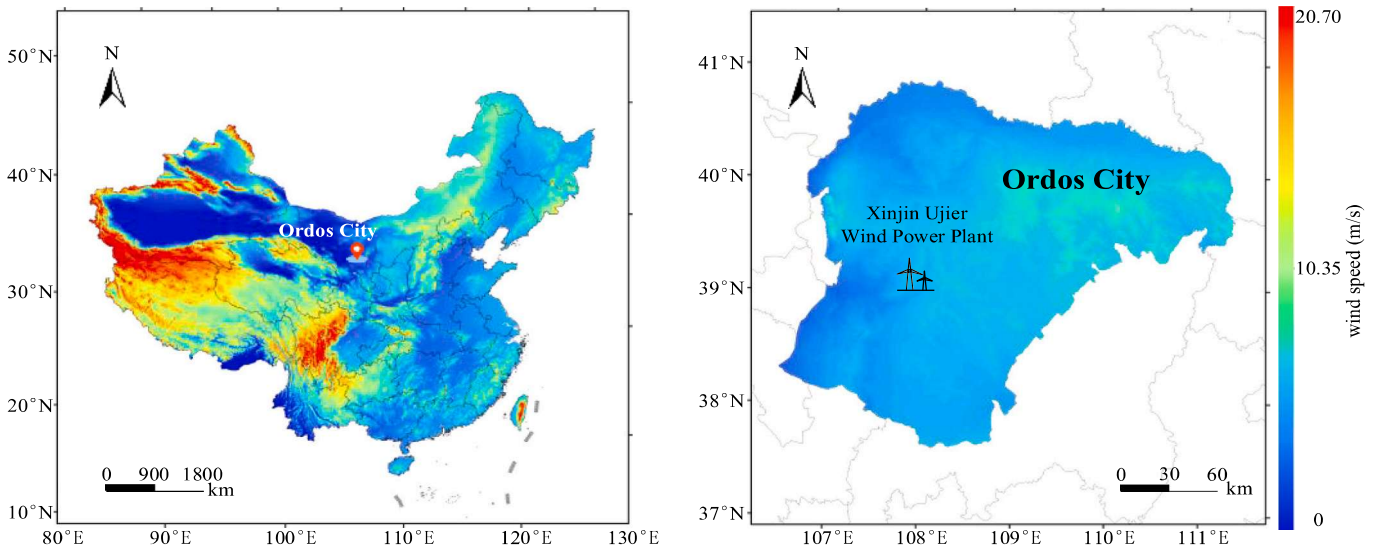


Fig. 8. Geographical Location of the Xinjin Ujier Wind Farm in Ordos, China.

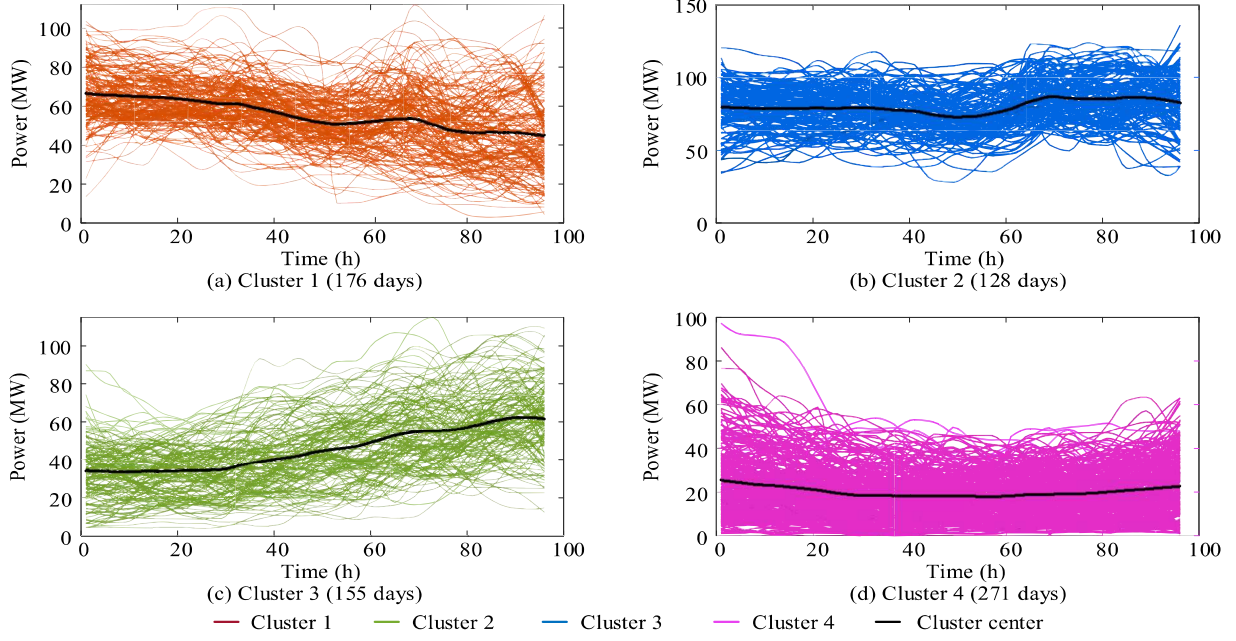


Fig. 9. Cluster-Based Power Distribution and Centroid Identification.

independent variable of the generated wind power sequence, $\mathcal{H}$ is the bandwidth of the KDE, $\hat{l}_n^c$ is the wind power sequence at the n -th scenario, and $V(\cdot)$ is the kernel function used in the KDE. In this study, we choose the Epanechnikov kernel, which has lower computational requirements, for the estimation, as shown in Eq. (11).

$$V(\pi) = \frac{3}{4}(1 - \pi^2) \quad (11)$$

where π is the independent variable of the kernel function. Based on the operation of $\hat{f}(\hat{l}^c)$, the Cumulative Distribution Function (CDF) $\hat{F}(\hat{l}^c)$ is obtained, and the quantile value ψ_{δ_j} at the quantile level δ_j is computed.

This forms the prediction interval γ at the desired confidence level P_γ , expressed as follows:

$$\psi_{\delta_j} = \hat{F}(\delta_j)^{-1} \quad (12)$$

$$P_\gamma = [\psi_{\delta_j}, \psi_{\bar{\delta}_j}] \quad (13)$$

$$\gamma_j = \bar{\delta}_j - \delta_j. \quad (14)$$

where $\bar{\delta}_j$ and δ_j are the upper and lower quantile levels of P_γ , j is the quantile marker.

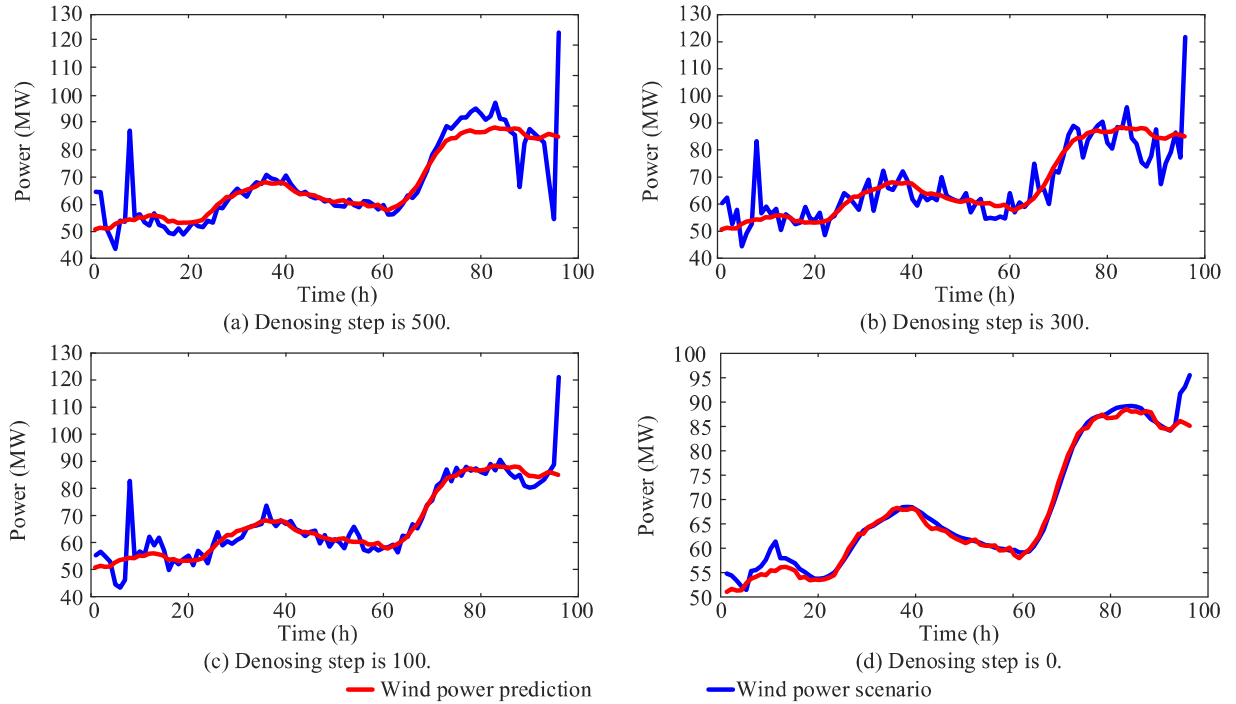


Fig. 10. Training loss convergence of the conditional diffusion model across clusters.

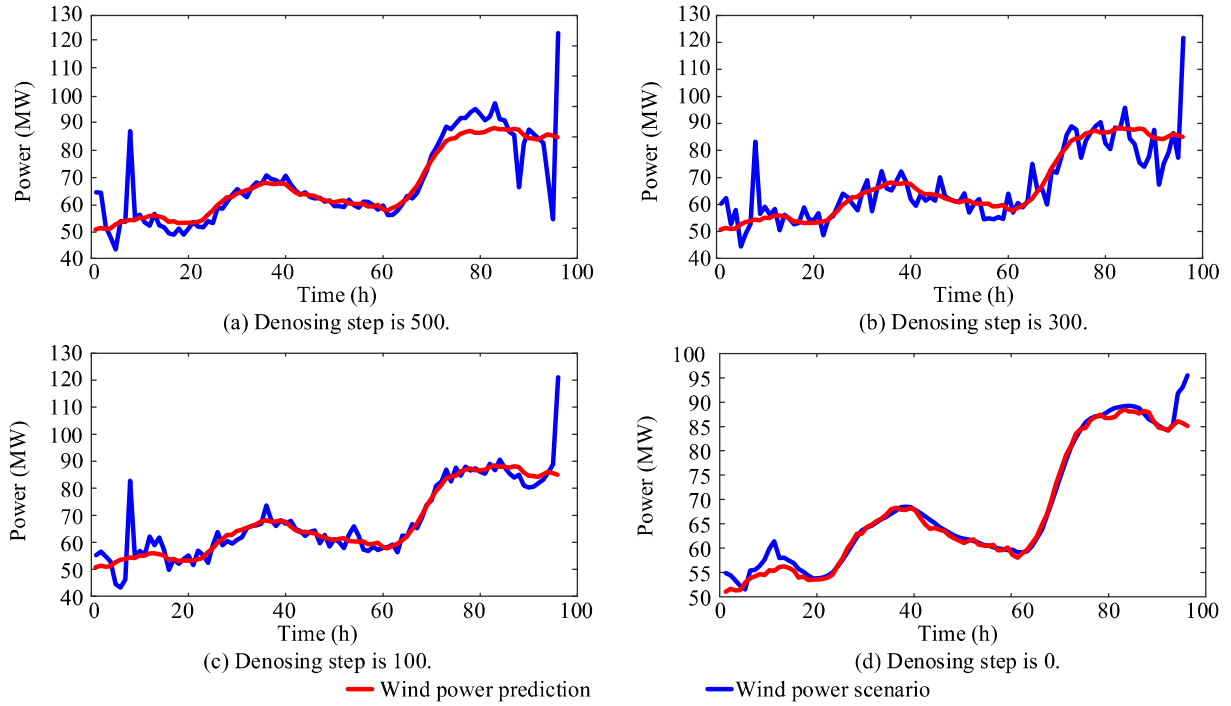


Fig. 11. Stepwise denoising process in wind power scenario generation.

6. Case study

To assess the reliability and effectiveness of the proposed probabilistic forecasting framework, this study selects the Xinjin Ujier Wind Farm in the Ordos region of Inner Mongolia, China, with an installed capacity of 149.5 MW, as the experimental site. The geographical location of the wind farm is shown in Fig. 8. The experimental dataset consists of actual wind power values, forecasted wind power values, and associated meteorological features recorded from January 1, 2020, to December 31, 2021, with a 15-min sampling interval, forming the training set. The test set, following the same structure, covers the period from January 1, 2022, to December 31, 2022. The meteorological features include wind speed, wind direction, temperature, atmospheric pressure, and relative humidity. Among these, the three most relevant variables—wind speed, temperature, and atmospheric pressure—are selected using the MIC method. All data were sourced from the wind farm's Supervisory Control and Data Acquisition (SCADA) system. To optimize the K-Medoids clustering process, the Davies-Bouldin index, Silhouette coefficient, and Elbow method are used, determining the optimal number of clusters to be four. The data sizes for Wind Clusters 1 to 4 are 96×176 , 96×128 , 96×155 , and 96×271 , respectively. The power distributions of each cluster and their centroids are illustrated in Fig. 9.

While the centroids may appear to exhibit lower variability than some individual wind power scenarios, they effectively represent the most central and structurally typical curves within each cluster. This is a direct result of the K-Medoids algorithm, which selects actual samples that minimize the total distance to all other members in the cluster,

ensuring physical realism and structural consistency.

The probabilistic forecasting experiments are implemented in MATLAB 2023b, utilizing the Deep Learning Toolbox, Parallel Computing Toolbox, and Statistics and Machine Learning Toolbox for model design and training. The experiments are conducted on a PC equipped with an Intel Core i7-10875H CPU and 16GB RAM. The diffusion model is trained with 500 diffusion time steps, using a noise schedule ranging from 10^{-4} to 10^{-3} . The Adam optimizer is applied with a batch size of 32 and an initial learning rate of 10^{-3} , which is later fine-tuned between 10^{-4} and 10^{-5} to enhance model performance.

6.1. Iterative process analysis

The variation in the loss function of the conditional diffusion model during training for each wind power cluster is shown in Fig. 10. The results indicate that after a limited number of training epochs, the models for each wind power cluster gradually converge. For smaller clusters, such as Cluster 2 and Cluster 3, convergence occurs more rapidly with lower final loss values due to their relatively constrained feature spaces. In contrast, larger clusters, such as Cluster 4, present more complex conditional distributions, requiring the model to learn a broader feature space, which leads to a slower loss reduction and a longer convergence process. Cluster 1, with a moderate data size, exhibits a stable loss curve throughout training. Notably, the introduction of the SCMM allows a single conditional diffusion model to handle the forecasting task effectively, significantly improving training efficiency.

Fig. 11 illustrates the denoising process applied to the Wind Cluster 2 dataset, using a randomly selected day for simplicity. Unlike in image generation tasks, wind power curves do not require complete transformation into Gaussian noise during the forward diffusion process. As shown in Fig. 9(a), the initial curve retains key temporal characteristics of the original data, preventing the model from drifting into unrealistic regions during denoising. This design ensures that the generated wind power curves closely align with real-world fluctuations. Furthermore, the gradual transformation from Fig. 11(a) to Fig. 11(d) demonstrates how the model progressively refines the forecast, effectively capturing both trend and uncertainty. The final generated results closely follow the

Table 1
Fast-DTW distance computation results for seasonal wind power forecasting.

Season	Fast-DTW Distance /MW			
	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Spring	1526.3078	3919.5276	561.3539	1823.6211
Summer	3133.3057	1844.3581	4175.1709	6696.3688
Autumn	2269.2549	680.0128	3475.1521	5886.2058
Winter	2798.8851	1233.1981	3248.4964	6032.4664

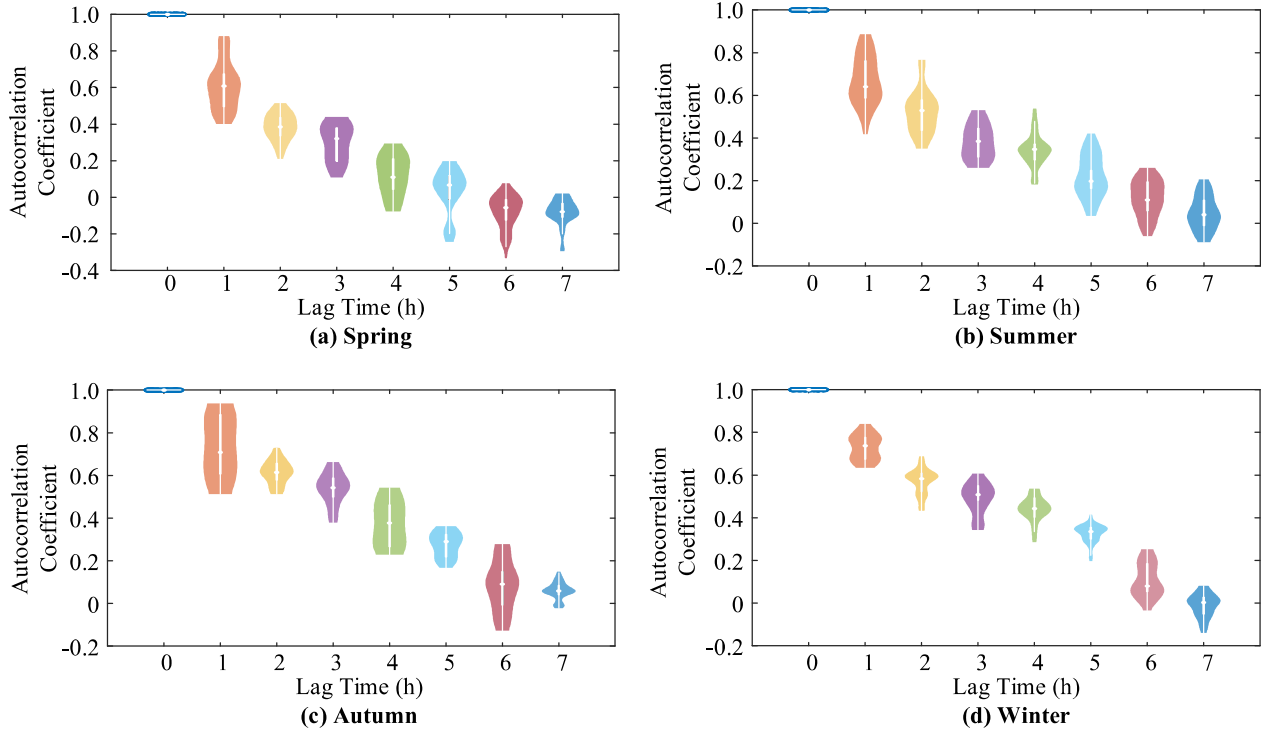


Fig. 12. Autocorrelation coefficients of generated wind power scenarios.

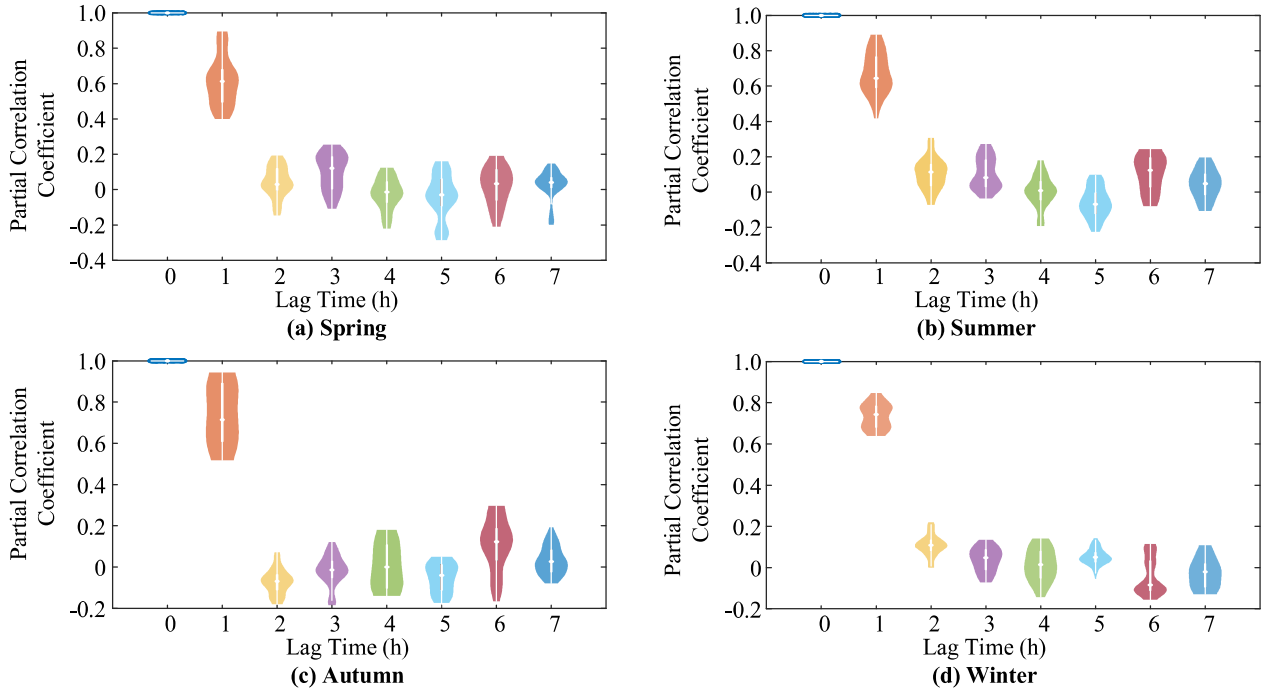


Fig. 13. Partial autocorrelation analysis of generated wind power scenarios.

predicted wind power values, maintaining a consistent variation pattern and enhancing forecasting accuracy.

6.2. Comparison and analysis of probability prediction results

6.2.1. Probability prediction result analysis

To evaluate the effectiveness of the proposed probabilistic forecasting framework in quantifying wind power uncertainty, typical daily

data from spring, summer, autumn, and winter are analyzed from the test set. Table 1 presents the wind power forecasts for each season along with the Fast-DTW distances to the cluster centroids. The optimal conditional diffusion model for each seasonal case is determined by selecting the cluster with the minimum distance. Using the chosen model, 100 wind power scenarios are generated, and the PDF is fitted to estimate prediction intervals at various confidence levels.

Figs. 12 and 13 illustrate the violin plots of the autocorrelation

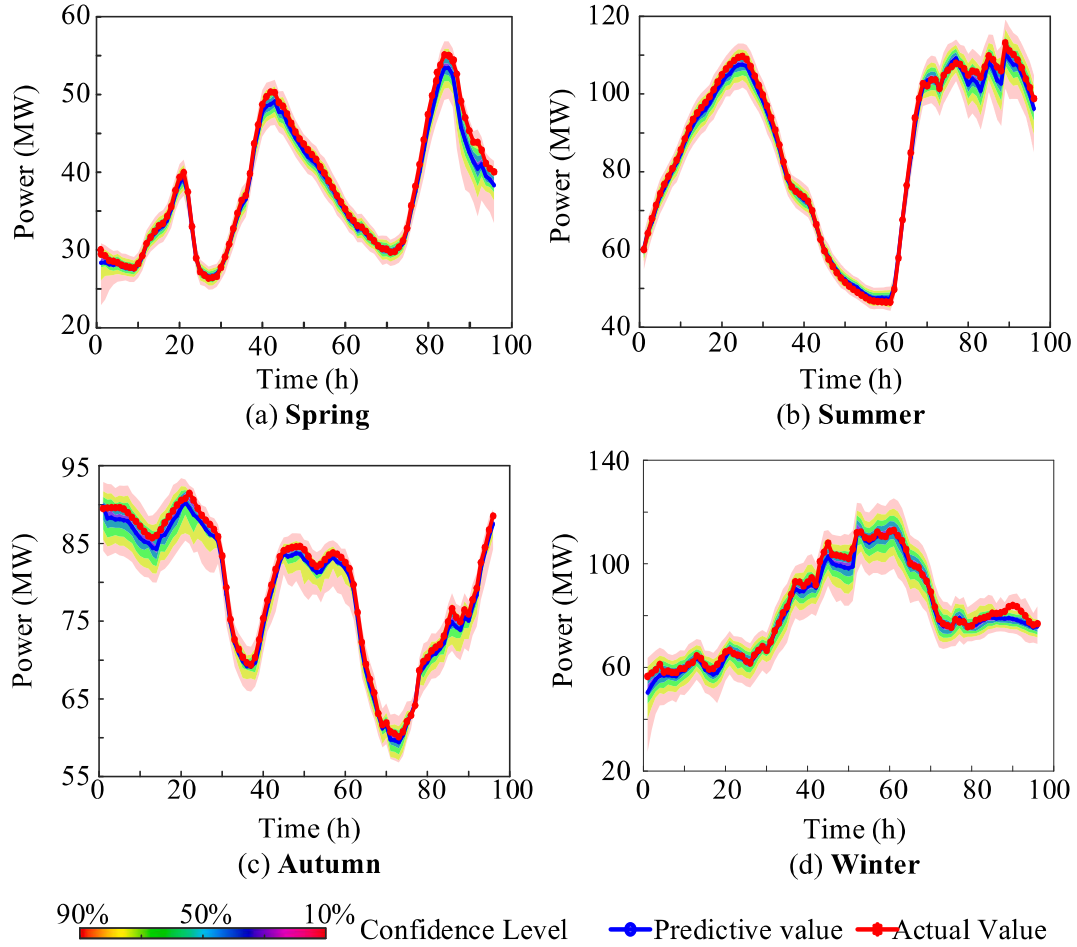


Fig. 14. Wind power probability prediction intervals across multiple confidence levels.

coefficients and partial autocorrelation coefficients for the generated wind power scenarios. In these plots, the circles represent the median values, while the width indicates the frequency distribution of the data. Fig. 12 shows that the generated scenarios exhibit strong autocorrelation within lag times of 1 to 7 h. The coefficient gradually decreases as the time interval increases, highlighting a strong short-term dependency and consistent variations in wind power output over short periods.

Fig. 13 further reveals that the partial autocorrelation coefficient peaks at a 1-h lag, indicating a strong direct relationship between the current wind power output and that of the previous hour. As the time interval increases beyond 1 h, the partial correlation weakens, suggesting a diminishing direct relationship between consecutive outputs over time. These results confirm that the generated wind power scenarios effectively capture the temporal dependencies of actual wind power output while maintaining consistency with its statistical properties.

Fig. 14 illustrates the probability prediction intervals of wind power at different confidence levels (90 %, 70 %, 50 %, 30 %, 10 %) alongside actual and predicted values. The findings show that the forecast intervals effectively encompass actual and forecast values and trace their trends of variation closely. During periods of rapid power changes, for instance, near the 30-h and 70-h marks during spring and autumn, the forecast intervals widen moderately, effectively encompassing the range of variations in actual values. Conversely, if power is not constant, for instance, at night during summer and winter, the gaps are quite small, reflecting high sparseness and accuracy of prediction.

At peak and trough, that is, in winter and autumn, prediction intervals increase somewhat to account for the additional uncertainty created by extreme fluctuations. However, intervals remain close

enough without excessive expansion. On average, probabilistic forecasting approach dynamically varies the sizes of intervals based on fluctuation in wind power such that full coverage of true and predicted values is ensured, but uncertainty is properly accounted for. These results highlight the usability and practical application of the model.

6.2.2. Comparison of probability prediction results

For further cross-validation of the proposed model, comparisons were drawn with Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP), VAE, and NF using the seasonal test cases described in Section 6.2.1. The models were compared against three significant metrics: Prediction Interval Coverage Probability (PICP), Prediction Interval Normalized Average Width (PINAW), and Comprehensive Weighted Coverage (CWC). PICP measures the percentage of actual values falling within the prediction interval, with higher values indicating better performance. PINAW evaluates the compactness of the interval, where lower values are preferred. CWC balances coverage and interval width, with a lower value representing superior overall performance. The mathematical formulations for these metrics can be found in [46].

The results of the probabilistic prediction intervals are depicted in Fig. 15, while the numerical comparisons of model performance are summarized in Table 2.

Based on the analysis of Fig. 15 and the data in Table 2, the following conclusions can be drawn:

- 1) **PICP Metric Analysis:** The proposed model consistently achieves higher coverage rates compared to baseline models across all confidence levels. In autumn, it attains PICP values of 0.9610, 0.9305, and

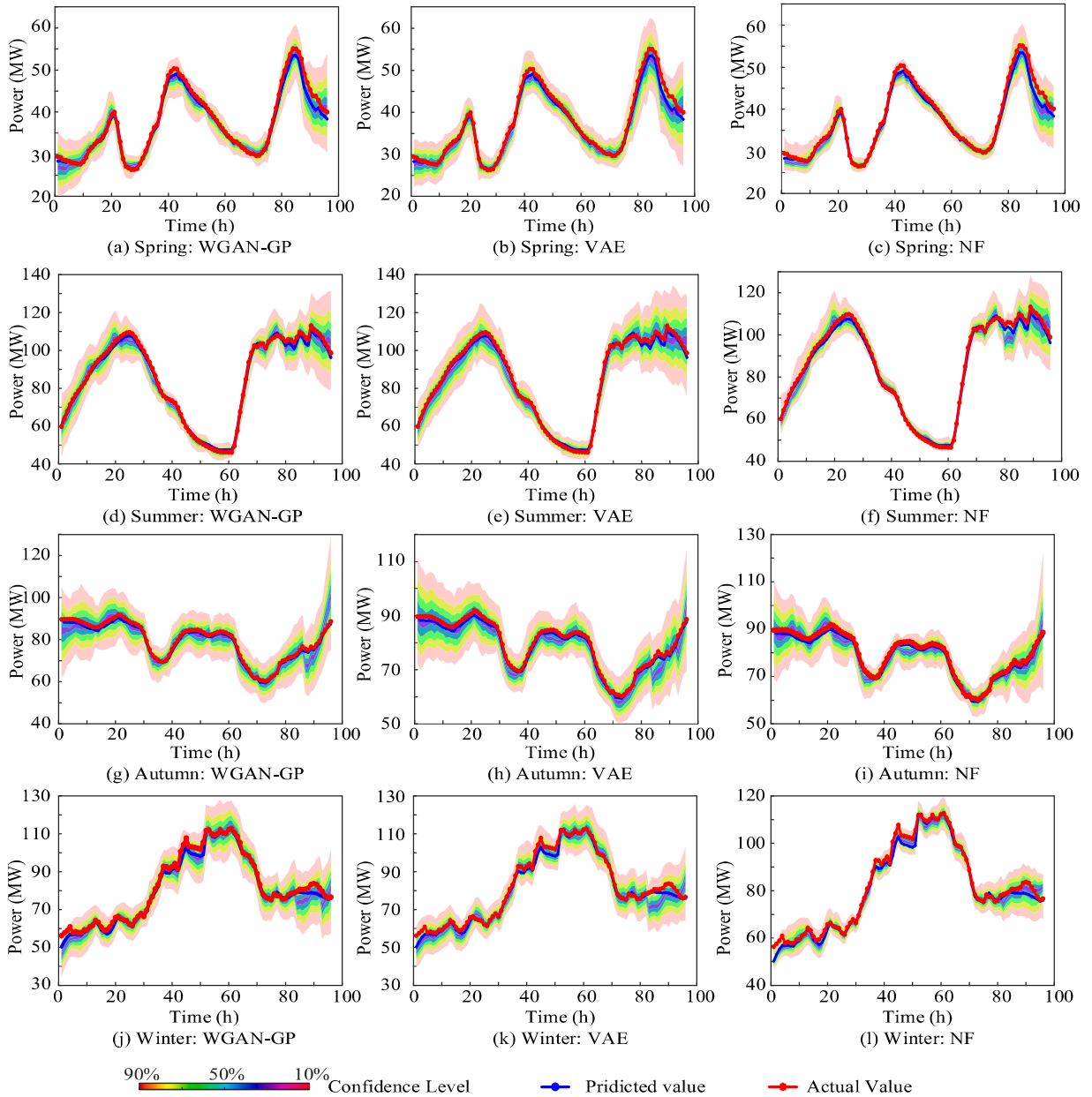


Fig. 15. Analysis of probability prediction intervals among different models.

0.8623 at 90 %, 85 %, and 80 % confidence levels, respectively, outperforming WGAN-GP (0.9296, 0.8615, 0.8089), VAE (0.9273, 0.8563, 0.7953), and NF (0.9314, 0.887, 0.8255). Compared to WGAN-GP, coverage improves by 3.38 %, 7.79 %, and 6.34 % at the corresponding confidence levels. Against VAE, the increases are 3.63 %, 9.83 %, and 10.31 %, while compared to NF, the improvements are 1.29 %, 3.38 %, and 3.48 %. These results demonstrate the model's superior coverage stability, particularly at lower confidence levels, confirming its robustness in capturing wind power uncertainty.

2) **PINAW Metric Analysis:** The proposed model produces significantly narrower prediction intervals than competing methods. In winter, its PINAW values at 90 %, 85 %, and 80 % confidence levels are 0.1077, 0.0954, and 0.0857, respectively, which are substantially lower than those of WGAN-GP (0.2167, 0.1990, 0.1729), VAE (0.2366, 0.2090, 0.1788), and NF (0.1677, 0.1491, 0.1143). Compared to WGAN-GP, the PINAW values decrease by 50.30 %, 52.06 %, and 50.45 % at the respective confidence levels. Reductions

compared to VAE are 54.46 %, 54.35 %, and 52.05 %, while those against NF are 35.78 %, 35.99 %, and 25.03 %. These results confirm that the proposed model not only minimizes prediction interval width but also significantly enhances forecasting accuracy.

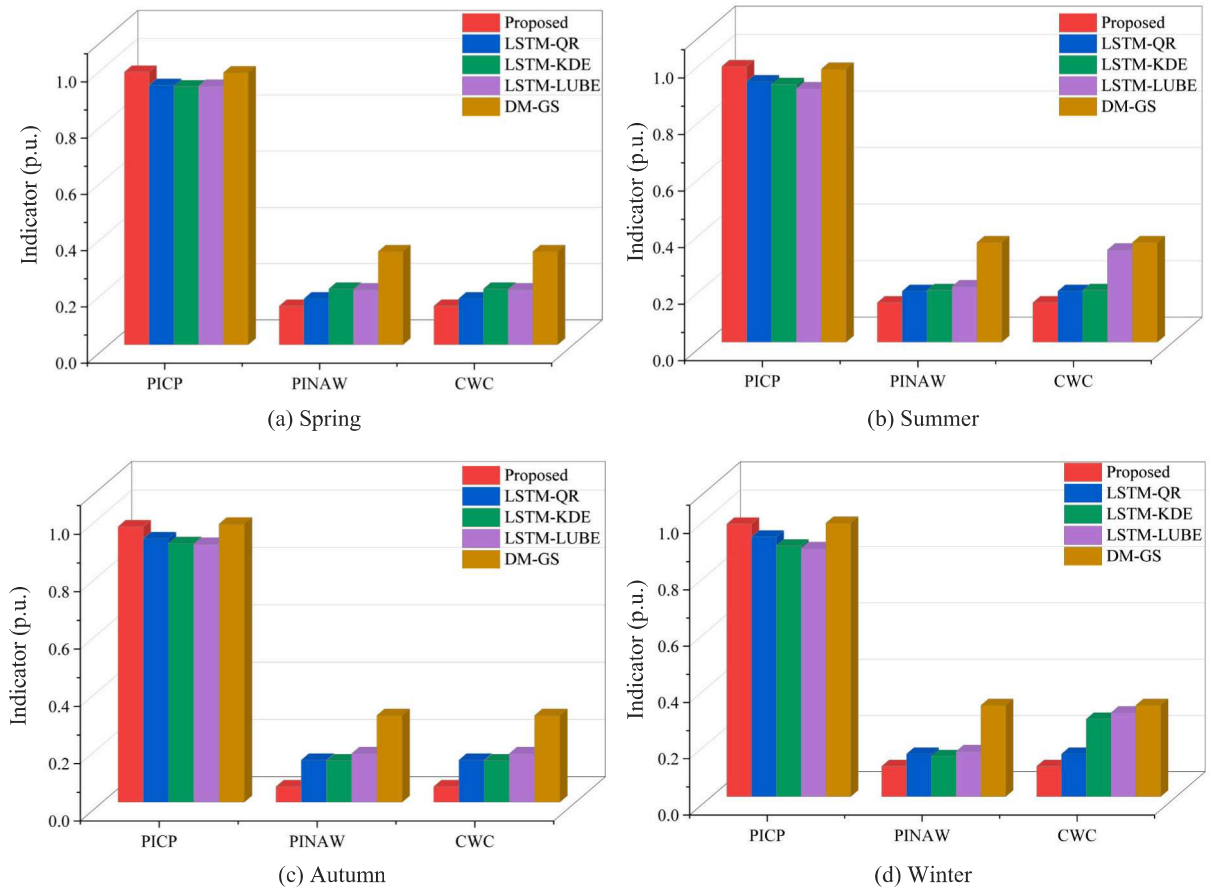
3) **CWC Performance Score Comparison:** The proposed model shows a tremendous performance score improvement in CWC. At the 80 % confidence level during spring, its CWC score is 0.1099, significantly lower than that of WGAN-GP (2.0637) and VAE (3.4861), representing a decrease by 94.67 % and 96.85 %, respectively. This improvement is attributed primarily to the model's high PICP values, which are penalty-free for under-coverage, and offer a strong balance between CWC and PINAW. Such outcomes show that not only does the model effectively minimize prediction intervals but also maintains improved prediction accuracy, favoring its overall performance advantage.

Briefly, compared to baseline generative models, the proposed framework is more covered with narrower predictive intervals, showing

Table 2

Performance comparison of probability prediction metrics across forecasting models.

Model	Confidence Interval	I_{PICP}				I_{PINAW}				I_{CWC}			
		Spring	Summer	Autumn	Winter	Spring	Summer	Autumn	Winter	Spring	Summer	Autumn	Winter
Proposed	90 %	0.9696	0.9751	0.9610	0.9681	0.1378	0.1399	0.0541	0.1077	0.1378	0.1399	0.0541	0.1077
	85 %	0.9401	0.9423	0.9305	0.9399	0.1220	0.1254	0.048	0.0954	0.1220	0.1254	0.048	0.0954
	80 %	0.898	0.8891	0.8623	0.8793	0.1099	0.1137	0.0427	0.0857	0.1099	0.1137	0.0427	0.0857
WGAN-GP	90 %	0.9268	0.9215	0.9296	0.9273	0.1881	0.1810	0.2318	0.2167	0.1881	0.1810	0.2318	0.2167
	85 %	0.8577	0.8632	0.8615	0.8486	0.1742	0.1689	0.2002	0.1990	0.1742	0.1689	0.2002	1.3333
	80 %	0.7927	0.8109	0.8089	0.7865	0.1347	0.1234	0.1655	0.1729	2.0637	0.1234	0.1655	3.5431
VAE	90 %	0.9312	0.9396	0.9273	0.9316	0.2132	0.2252	0.2796	0.2366	0.2132	0.2252	0.2796	0.2366
	85 %	0.8516	0.8472	0.8563	0.8530	0.1517	0.1396	0.1834	0.2090	0.1517	1.4262	0.1834	0.2090
	80 %	0.7866	0.7817	0.7953	0.8132	0.1460	0.1224	0.1620	0.1788	3.4861	5.3137	1.6885	0.1788
NF	90 %	0.9489	0.9488	0.9314	0.9501	0.1632	0.1679	0.2441	0.1677	0.1632	0.1679	0.2441	0.1677
	85 %	0.9076	0.9001	0.887	0.9084	0.1443	0.1393	0.1745	0.1491	0.1443	0.1393	0.1745	0.1491
	80 %	0.8376	0.8333	0.8255	0.8515	0.1244	0.1308	0.1515	0.1143	0.1244	0.1308	0.1520	0.1143

**Fig. 16.** Comparison of models under different probability estimation methods.

better performance in probabilistic wind power forecasting. Furthermore, we conducted a systematic comparison of the probability estimation methods employed by the framework. The comparative models included the Long Short-Term Memory (LSTM)-based Quantile Regression (LSTM-QR), the LSTM-based KDE (LSTM-KDE), the LSTM-based Lower and Upper Bound Interval Estimation (LSTM-LUBE), and the enhanced diffusion model that integrates Gaussian distribution estimation (DM-GS). Notably, the training and testing datasets for the LSTM models remained consistent with those used in the proposed model.

Fig. 16 presents a comparison of the probabilistic forecasting performance of the different models at the 90 % confidence level. The results at other confidence levels follow a similar trend and, therefore, are omitted for brevity.

As shown in **Fig. 16**, compared to the DM-GS method, which assumes a parametric distribution, the proposed model in this study performs slightly worse in terms of the PICP. However, it achieves a significant reduction in the PINAW. It is important to note that the higher confidence coverage level of the DM-GS method is achieved at the expense of

Table 3

Impact of key framework components based on ablation study results.

Model	Loss				
	Spring	Summer	Autumn	Winter	Mean
Proposed	0.03544	0.03976	0.03887	0.04024	0.03858
E1	0.06453	0.08479	0.05547	0.08712	0.07298
E2	0.04016	0.04503	0.04371	0.04819	0.04427
E3	0.1035	0.1424	0.13028	0.1517	0.1320
E4	0.1627	0.2443	0.2197	0.2516	0.2196

a wider interval width. In practical applications, excessively wide prediction intervals often fail to provide meaningful guidance for decision-making. Further comparison with LSTM-based non-parametric estimation methods reveals that, while ensuring a relatively superior PICP, the proposed model effectively narrows the PINAW. Notably, when compared with the LSTM-KDE and LSTM-LUBE methods, these two approaches exhibit distinct peaks in the composite indicator CWC during the summer and winter, which reflects their failure to meet the preset confidence level during these periods, leading to poorer prediction accuracy. The analysis indicates that the non-parametric estimation method used in the proposed model demonstrates better envelope properties and is able to more comprehensively and precisely capture the uncertainty characteristics of wind power generation.

6.3. Ablation experiment

To examine the role of every element in the setup of the considered wind power probabilistic forecasting, a series of ablation experiments were conducted. By removing important elements one by one in an organized manner, their impact on model performance was quantified using the loss function across four test sets for spring, summer, autumn, and winter. In a substitution check, replacing EMD with STL or a discrete wavelet decomposition resulted in looser prediction intervals at comparable coverage, indicating that EMD with entropy selection provides stronger envelope quality for this dataset. Results are presented in Table 3. The ablation configurations of all experiments are presented

below:

E1: Removes the Conditional Network module and sends the sequence directly as conditional input to the Denoising Network.

E2: Removes the Adapter Module and directly passes condition network information to the Denoising Network.

E3: Deletes the SCMM, training the diffusion model without dividing a special training set.

E4: At the same time removes the Conditional Network module and the SCMM.

Results of Table 3 indicate that each and every aspect of the model is responsible for its efficiency. Removal of the Conditional Network module from E1 has the average loss increase by 89.17 %, illustrating just how essential this module is to offer dynamic conditional knowledge to the denoising process. For E2, the removal of the Adapter Module results in its effect as a 14.75 % increase in average loss, suggesting that even though its impact is lesser, it also contributes indirectly to increasing the predictive power of the model. The maximum decrease in performance in E3 is caused by the removal of SCMM, resulting in a 242.15 % increase in average loss. SCMM is critical to finding and catching wind power patterns, homogenizing the training sets, and improving denoising efficiency. Its removal makes the model's ability to effectively learn temporal dependencies unstable, causing severe performance deterioration. The worst-case scenario is achieved in E4 when SCMM and Conditional Network are removed, causing a 469.21 % increase in average loss, which confirms that they are essentials for maintaining model stability and accuracy.

In general, the Conditional Network module and SCMM are the most critical components of the framework and play a crucial role in improving the accuracy of forecasting and denoising. The effect of the Adapter Module is relatively smaller, but it also helps to improve the model's generalization ability across different datasets. The combined effect of all these modules allows the framework to more effectively capture complex temporal relationships and wind power variations, thereby ultimately leading to improved probabilistic forecasting performance.

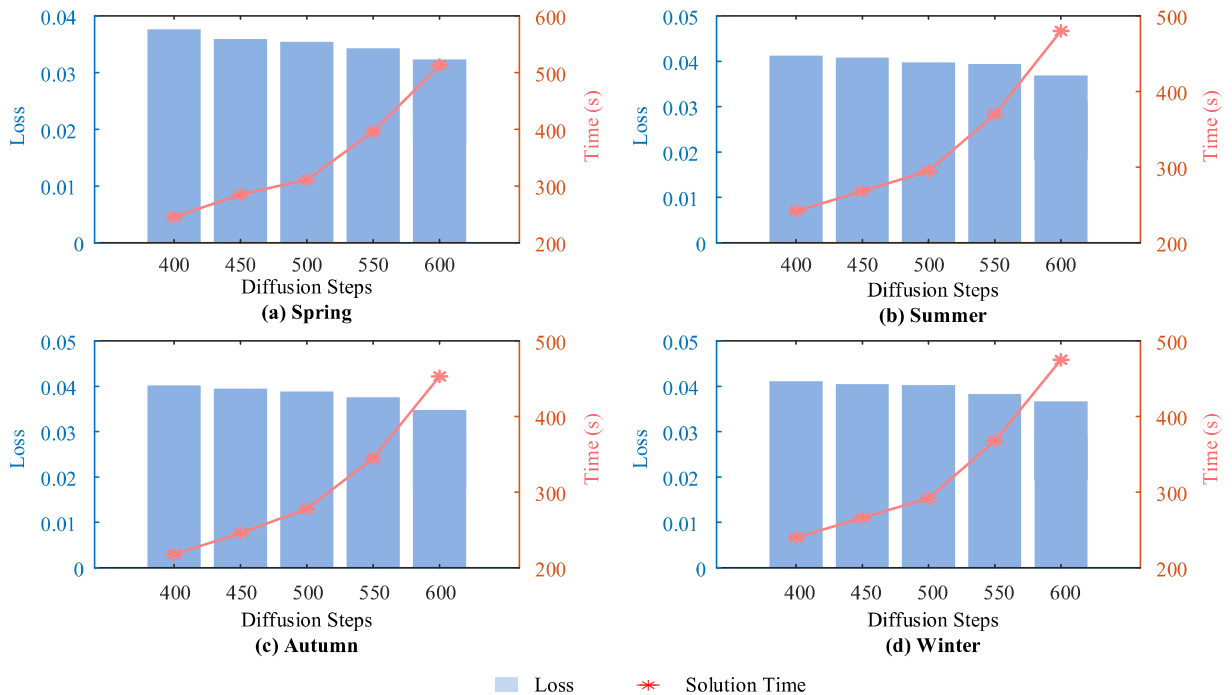


Fig. 17. Influence of diffusion timesteps on forecasting accuracy and computational efficiency.

Table 4

Probabilistic forecasting results of the proposed framework under different dataset sizes.

Dataset size	I_{PICP}			
	Spring	Summer	Autumn	Winter
10 %	0.7046	0.7301	0.7120	0.7272
40 %	0.8846	0.8722	0.8898	0.8826
70 %	0.9495	0.9503	0.9447	0.9532
100 %	0.9696	0.9751	0.9610	0.9681
110 %	0.9709	0.9755	0.9613	0.9753
120 %	0.9713	0.9760	0.9619	0.9763
130 %	0.9720	0.9762	0.9621	0.9766

6.4. Sensitivity analysis

6.4.1. Impact of diffusion step size on the model

The number of diffusion timesteps is a critical parameter that influences both model performance and computational efficiency in diffusion models. This section discusses the impact of varying the timestep size on processing time and forecast accuracy.

As shown in Fig. 17, increasing the diffusion timesteps from 400 to 600 leads to a significant improvement in model performance across all test sets, but also results in a substantial rise in computational time. Specifically:

- In spring, the loss decreased from 0.0376 to 0.0325 (a 13.56 % reduction), while computation time increased from 245.16 to 516.28 s (a 110.59 % increase).
- In summer, the loss decreased from 0.0412 to 0.0371 (a 9.95 % reduction), while computation time increased from 240.08 to 481.25 s (a 100.45 % increase).
- In autumn, the loss decreased from 0.0402 to 0.0352 (a 12.44 % reduction), while computation time increased from 218.37 to 454.49 s (a 108.13 % increase).

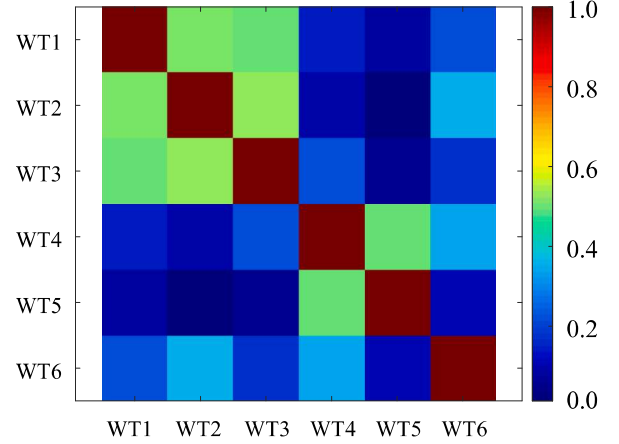


Fig. 19. Relationship between power datasets of different wind farms.

- In winter, the loss decreased from 0.0411 to 0.0370 (a 9.98 % reduction), while computation time increased from 241.34 to 478.27 s (a 98.17 % increase).

The results indicate that increasing the diffusion timesteps enhances prediction accuracy across all seasons, with the most significant improvement observed in spring, where the loss reduction reached 13.56 %. However, this improvement comes at the expense of computational efficiency, with processing time more than doubling in some cases.

In practical applications, selecting an appropriate timestep length requires balancing accuracy and efficiency. While longer timesteps yield better predictions, they also impose a higher computational cost. Therefore, an optimal timestep should be chosen based on specific application constraints to achieve the best trade-off between forecasting performance and processing time. This analysis also demonstrates that the solution times of the proposed framework remain acceptable, which

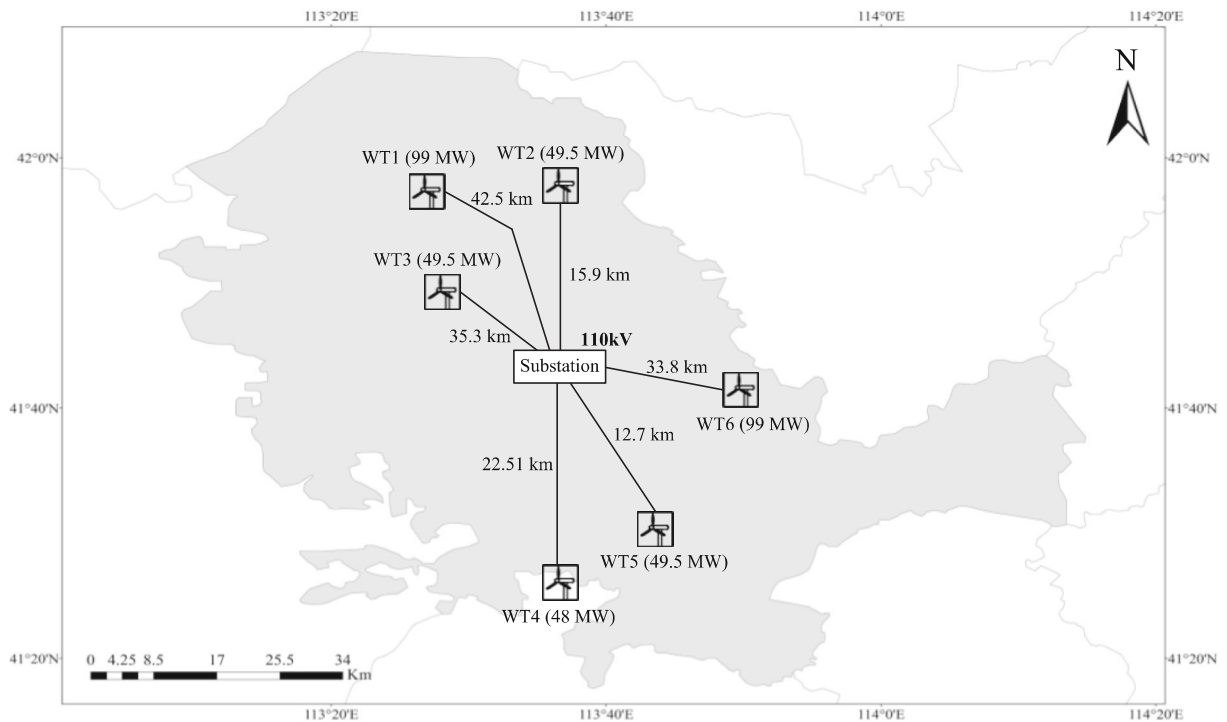


Fig. 18. Topological map of the wind farm cluster in a region of Inner Mongolia.

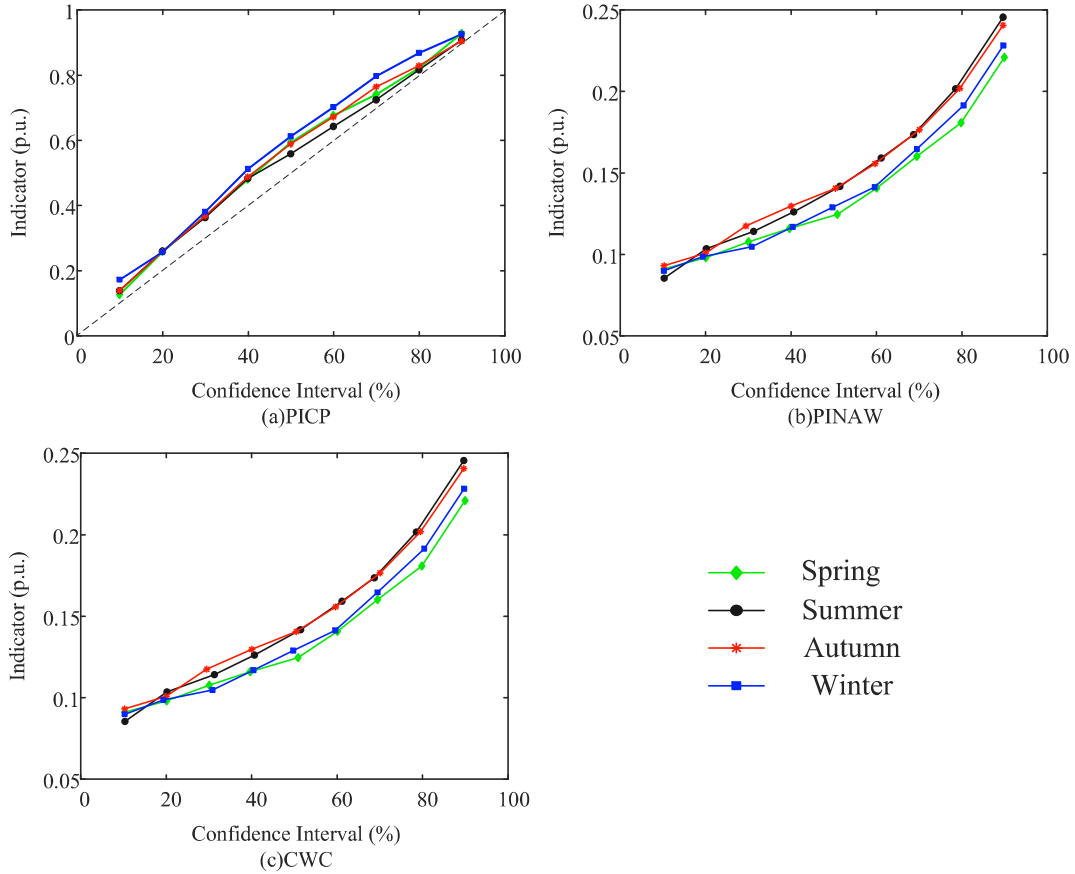


Fig. 20. Probabilistic forecasting of wind farm clusters' output.

further supports the feasibility of maintaining a limited number of diffusion models corresponding to different wind power regimes.

6.4.2. Impact of dataset size on the model

In this section, we investigate the impact of dataset size on the probabilistic forecasting performance. For each test, the original dataset is used at sizes of 10 %, 40 %, 70 %, 100 %, 110 %, 120 %, and 130 %. Table 4 presents the probabilistic forecasting results of the proposed framework under different dataset sizes, with a confidence level of 90 %.

As shown in Table 4, with the expansion of the dataset size, the model's probabilistic forecasting performance improves significantly. However, when the dataset reaches 100 % and beyond, the performance gain becomes more gradual. Specifically, the PICP value for the spring season increases from 0.7046 with a 10 % dataset to 0.9720 with a 130 % dataset. A similar trend is observed for other seasons: summer rises from 0.7301 to 0.9762, autumn from 0.7120 to 0.9621, and winter from 0.7272 to 0.9766. This analysis indicates that as the dataset size increases, the model becomes more capable of capturing local patterns within the wind power sequences, leading to a notable improvement in the accuracy and reliability of probabilistic forecasts. However, it is important to note that beyond 100 % dataset size, although larger datasets contribute to further reducing forecasting errors and enhancing model performance, the marginal benefits diminish. Moreover, this increase in dataset size also comes with significantly higher data acquisition and processing costs.

6.5. Validation of framework scalability

Due to the extensive geographic distribution and large number of regional wind farm clusters, the overall power output exhibits higher randomness and greater variability [49]. To further validate the

effectiveness and applicability of the proposed framework, this section selects a typical wind farm cluster in a region of Inner Mongolia as the research subject, as shown in Fig. 18. The cluster consists of six wind farms with installed capacities of 99 MW, 49.5 MW, 49.5 MW, 48 MW, 49.5 MW, and 99 MW, respectively. The method for obtaining the power data set is consistent with the approach described earlier. The Pearson correlation coefficients between the power data sets of different wind farms are shown in Fig. 19. The Pearson correlation coefficient effectively reflects the correlation between two variables, with a range of $[-1, 1]$. The closer the result is to 1, the higher the data correlation [50].

As shown in Fig. 19, during the data collection period, the Pearson correlation coefficients between the datasets of the various wind farms ranged from 0 to 0.6. These relatively low correlation coefficients indicate that the power generation characteristics of the six wind farms exhibit a complex relationship, with significant randomness and variability. This characteristic provides a solid testing foundation for subsequently validating the model's generalization ability and the forecasting capability of wind farm clusters. Next, the data from each wind farm across the spring, summer, autumn, and winter seasons were reconstructed to form a cluster dataset for probabilistic forecasting, with the results of various metrics presented in Fig. 19. Notably, compared to the method of individually predicting each wind farm's data and then aggregating the results, the maximum data range of the reconstructed cluster dataset increased by a factor of 2.3.

As shown in Fig. 20, with the increase in the confidence interval, the PICP value demonstrates a stable upward trend, and the PINAW also increases correspondingly, though it remains within an acceptable range overall. At the same time, at all confidence levels, the model's predictions consistently meet the preset standards, with the CWC values always aligning with the PINAW, without triggering any penalty terms. The analysis indicates that, despite the fact that the actual power

fluctuations of the wind farm cluster are more intense and frequent than those of a single large wind farm, the proposed forecasting framework is still able to provide relatively accurate probabilistic prediction intervals, demonstrating good robustness.

7. Conclusion and future work

This paper presents a novel wind power probabilistic forecasting framework that integrates the SCMM with an enhanced Conditional Diffusion Model to effectively quantify wind power uncertainty. A CDJN is embedded into the diffusion model, where a dynamically structured Conditional Network replaces traditional static encoding, significantly enhancing the model's ability to capture complex dependencies. To address the high heterogeneity and complexity of wind power curves, SCMM is introduced, leveraging K-Medoids clustering and Fast-DTW to construct dedicated training sets for each prediction date. These refined datasets enable the conditional diffusion model to generate realistic wind power scenarios, with KDE applied to produce probabilistic forecasts. The effectiveness of the proposed framework is validated through extensive case studies, leading to the following key conclusions:

- 1) The proposed Conditional Diffusion Model consistently outperforms traditional forecasting methods, demonstrating superior scenario coverage and compact prediction intervals across various confidence levels. Its advantage is particularly evident at lower confidence levels, indicating higher robustness and improved predictive accuracy in wind power probabilistic forecasting tasks.
- 2) The Conditional Network module and SCMM play a critical role in enhancing model performance. The ablation study reveals that removing these components results in a 469.21 % increase in average loss, confirming their importance in uncertainty quantification. Additionally, the Adaptive Module, while having a lesser impact, still contributes positively to overall predictive accuracy.
- 3) Increasing the diffusion timesteps improves forecasting accuracy, with a 13.56 % reduction in loss during spring. However, this comes at the cost of higher computational time, which increases by 110.59 %. Therefore, in practical applications, an optimal timestep length must be selected to balance computational efficiency and forecasting precision.
- 4) The proposed predictive framework remains capable of providing accurate probabilistic forecast intervals even in wind farm clusters with highly volatile and frequent fluctuations. This demonstrates its robust performance, effectively meeting the demands of practical engineering applications.

Future work will explore the integration of powerful predictive models with the probabilistic framework presented in this study, particularly for modeling extreme fluctuations in wind power generation. This will involve the incorporation of rare event detection and adaptive diffusion processes. Addressing these extreme cases is crucial for enhancing grid resilience and integrating wind power more effectively into power systems. Enhancing the model's ability to capture these extreme variations will be a key direction for further advancements in probabilistic wind power forecasting. A detailed investigation of day-ahead prediction models will also be addressed in future work.

CRedit authorship contribution statement

Yuan Gao: Writing – original draft, Software, Resources, Investigation, Funding acquisition, Formal analysis. **Sile Hu:** Supervision, Resources, Data curation. **Yahui Chen:** Writing – review & editing, Writing – original draft. **Muhammad Farhan Khan:** Software, Methodology, Data curation. **Xiaolei Cheng:** Supervision, Software, Data curation. **Jiaqiang Yang:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the DaqingshanLab (2024KYPT0011).

Data availability

Data will be made available on request.

References

- [1] Chen F, Zhang W, Chen R, et al. Adapting carbon neutrality: tailoring advanced emission strategies for developing countries. *Appl Energy* 2024;361:122845.
- [2] McCoy A, Musial W, Hammond R, et al. Offshore wind market report: 2024 edition. Golden, CO (United States): National Renewable Energy Laboratory (NREL); 2024.
- [3] Zhang Z, Shen X, Chang M. Dual stochastic multi-stage robust optimization of integrated energy system considering long, medium, and short-term uncertainties. *Appl Energy* 2025;400:126573.
- [4] Norouzi M, Aghaei J, Niknam T, et al. Risk-averse and flexi-intelligent scheduling of microgrids based on hybrid Boltzmann machines and cascade neural network forecasting. *Appl Energy* 2023;348:121573.
- [5] Xu B, Luan W, Yang J, et al. Integrated three-stage decentralized scheduling for virtual power plants: a model-assisted multi-agent reinforcement learning method. *Appl Energy* 2024;376:123985.
- [6] Yi P, Bao Z, Huang F, et al. Towards effective long-term wind power forecasting: a deep conditional generative spatio-temporal approach. *IEEE Trans Knowl Data Eng* 2024;36(12):9403–17.
- [7] Yin L, Zhang B, et al. Relaxed deep generative adversarial networks for real-time economic smart generation dispatch and control of integrated energy systems. *Appl Energy* 2023;330:120300.
- [8] Li D, Zhang Y, Yang Z, et al. Sensing anomaly of photovoltaic systems with sequential conditional variational autoencoder. *Appl Energy* 2024;353:122124.
- [9] Patil SR, Kulkarni R, Ghode SB, et al. Plasma-assisted surface engineering of binary metal chalcogenides: a path toward high energy efficiency, electrocatalysts for water splitting, and urea oxidation with stability prediction via machine learning. *ACS Appl. Energy Mater.* 2025;8(4):2346–59.
- [10] Li S, Xiong H, Chen Y, et al. DiffCharge: generating EV charging scenarios via a denoising diffusion model. *IEEE Trans Smart Grid*. 2024;15(4):3936–49.
- [11] Wang Z, Zhang H, et al. Customized load profiles synthesis for electricity customers based on conditional diffusion models. *IEEE Trans Smart Grid* 2024;15(4):4259–70.
- [12] Zhang L, Jiang Z, Ji T, et al. Diffusion-based inpainting approach for multifunctional short-term load forecasting. *Appl Energy* 2025;377:124442.
- [13] Song C, Yang H, Cai J, et al. Multi-energy load forecasting via hierarchical multi-task learning and spatiotemporal attention. *Appl Energy* 2024;373:123788.
- [14] Zhang R, Xue D, Wang Y, et al. Towards dense and accurate radar perception via efficient cross-modal diffusion model. *IEEE Robot Autom Lett* 2024;9(9):7429–36.
- [15] Yang L, Zhang Z, Song Y, et al. Diffusion models: a comprehensive survey of methods and applications. *ACM Comput Surv* 2023;56(4):1–39.
- [16] Xue Y, Xie X, Marin R, et al. Human 3Diffusion: Realistic avatar creation via explicit 3D consistent diffusion models. *arXiv preprint*; 2024. <https://doi.org/10.48550/arXiv.2406.08475>. *arXiv:2406.08475*.
- [17] Kumar K, Prabhakar P, Verma A, et al. Advancements in wind power forecasting: a comprehensive review of artificial intelligence-based approaches. *Multimed Tools Appl* 2024;1–30.
- [18] Jin H, Shi L, Chen X, et al. Probabilistic wind power forecasting using selective ensemble of finite mixture Gaussian process regression models. *Renew Energy* 2021;174:1–8.
- [19] Bounnite MY. Markovian modeling of wind energy systems. *Proc Int Workshop Math Model Renew Energy* 2024;1–6.
- [20] Sivhugwana KS, Ranganai E, et al. An ensemble approach to short-term wind speed predictions using stochastic methods, wavelets and gradient boosting decision trees. *Wind* 2024;4(1):3.
- [21] Moradian S, Gharbia S, Iglesias G, et al. A copula post-processing method for wind power projections under climate change. *Energy Convers Manag X* 2024;23:100660.
- [22] Gao Y, Tahir M, Siano P, et al. Optimization of renewable energy-based integrated energy systems: a three-stage stochastic robust model. *Appl Energy* 2025;377:124635.
- [23] Li Y, Wang B, Yang Z, et al. Hierarchical stochastic scheduling of multi-community integrated energy systems in uncertain environments via Stackelberg game. *Appl Energy* 2022;308:118392.
- [24] Dias JA. Long-term hourly scenario generation for correlated wind and solar power combining variational autoencoders with radial basis function kernels. *arXiv preprint*; 2023. <https://doi.org/10.48550/arXiv.2306.16427>. *arXiv:2306.16427*.

- [25] Harrou F, Dairi A, Dorbane A, et al. Enhancing wind power prediction with self-attentive variational autoencoders: a comparative study. *Results Eng* 2024;23: 102504.
- [26] Dumas J, Wehenkel A, Lanaspese D, et al. A deep generative model for probabilistic energy forecasting in power systems: normalizing flows. *Appl Energy* 2022;305:117871.
- [27] Liu Z, Guo H, Zhang Y, et al. A comprehensive review of wind power prediction based on machine learning: models, applications, and challenges. *Energies* 2025;18 (2):350.
- [28] Cao H, Tan C, Gao Z, et al. A survey on generative diffusion models. *IEEE Trans Knowl Data Eng* 2024;36(7):2814–30.
- [29] Liu H, Chen Z, Yuan Y, et al. AudioLDM: Text-to-audio generation with latent diffusion models. *arXiv preprint*; 2023. <https://doi.org/10.48550/arXiv.2301.12503>. arXiv:2301.12503.
- [30] Kim J, Song K, Lee G, et al. Time-series data clustering with load-shape preservation for identifying residential energy consumption behaviors. *Energ Buildings* 2024;311:114130.
- [31] Merizzi F, Asperti A, Colamonaco S, et al. Wind speed super-resolution and validation: from ERA5 to CERRA via diffusion models. *Neural Comput Appl* 2024; 36(34):21899–921.
- [32] Meng Q, Xu J, Ge L, et al. Economic optimization operation approach of integrated energy system considering wind power consumption and flexible load regulation. *J Electr Eng Technol* 2024;19(1):209–21.
- [33] Croitoru F-A, Hondru V, Ionescu RT, et al. Diffusion models in vision: a survey. *IEEE Trans Pattern Anal Mach Intell* 2023;45(9):10850–69.
- [34] Wang Y, Xu H, Zou R, Zhang F, Hu Q. Dynamic non-constraint ensemble model for probabilistic wind power and wind speed forecasting. *Renewable Sustain Energy Rev* 2024;204:114781.
- [35] Huang Y, Thammatadatrakoon A, Zheng W, et al. Probabilistic Gaussian superposition for efficient 3D occupancy prediction. *arXiv preprint*; 2024. arXiv: 2412.04384.
- [36] Li F, Zheng H, Li X, et al. A novel hybrid model for multi-step ahead photovoltaic power prediction based on conditional time series generative adversarial networks. *Renew Energy* 2022;199:560–86.
- [37] Yang T, Huang L, Fu P, et al. Combined multi-component composite time series power prediction model for distributed energy systems based on STL data decomposition. *Measurement* 2025;253:117299.
- [38] Shalby EM, Abdelaziz AY, Ahmed ES, et al. A comprehensive guide to selecting suitable wavelet decomposition level and functions in discrete wavelet transform for fault detection in distribution networks. *Sci Rep* 2025;15:1160.
- [39] Ober SW. Towards improved variational inference for deep Bayesian models. *arXiv preprint*; 2024. arXiv:2401.12418.
- [40] Pandey KK, Shukla D, et al. Min–max kurtosis mean distance based k-means initial centroid initialization method for big genomic data clustering. *Evol Intell* 2023;16 (3):1055–76.
- [41] Zhang Q, Zhang C, Cui L, et al. A method for measuring similarity of time series based on series decomposition and dynamic time warping. *Appl Intell* 2023;53(6): 6448–63.
- [42] Riyahi M, Gutiérrez Martín A. Optimizing capacity expansion modeling with a novel hierarchical clustering and systematic elbow method: a case study on power and storage units in Spain. *Energy* 2025;323:135788.
- [43] Yang M, Wang T, Zhang X, et al. Considering dynamic perception of fluctuation trend for long-foresight-term wind power prediction. *Energy* 2024;289:130016.
- [44] Yin Z-C, Wei J-J, Ying S, Jia P-N. Time-awareness in kernel density estimation for movement data. *Int J Geogr Inf Sci* 2025;39(7):1–18.
- [45] Nguyen N, Quanz B. Temporal latent auto-encoder: a method for probabilistic multivariate time series forecasting. *Proc AAAI Conf Artif Intell* 2021;35(10): 9117–25.
- [46] Gao Y, Zhao Y, Hu S, Tahir M, Yuan W, Yang J. A three-stage adjustable robust optimization framework for energy base leveraging transfer learning. *Energy* 2025; 319:135037.
- [47] Bai M, Zhou G, Yao P, et al. Deep multi-attribute spatial–temporal graph convolutional recurrent neural network-based multivariable spatial–temporal information fusion for short-term probabilistic forecast of multi-site photovoltaic power. *Expert Syst Appl* 2025;279:127458.
- [48] Gao W, Xiahou K, Liu Y, et al. Transient frequency-voltage support strategy for VSC-MTDC integrated offshore wind farms based on perturbation observer and funnel control. *IEEE Trans Sustain Energy* 2025;16(3):1931–43.
- [49] Xu X, Xiahou K, Du W, et al. Security assessment of cascading failures in cyber-physical power systems with wind power penetration. *IEEE Trans Power Syst* 2025; 1–13.
- [50] Wang Y, Wang J, Cao M, et al. Prediction method of wind farm power generation capacity based on feature clustering and correlation analysis. *Electr Pow Syst Res* 2022;212:108634. <https://doi.org/10.1016/j.epsr.2022.108634>.